



ScienceDirect

Chemical Engineering Journal

Date: 1 December 2025

[Show more](#) ✓[Research article](#) [Full text access](#) [Get rights and content](#) ↗

Multiscale interface engineering via femtosecond laser synergized with micro-Ag interlayer for robust ceramic-metal sintered-joining in protonic ceramic fuel cells

[Naiqiang Wang](#), [Xiaoqing Si](#) , [Hanyue Ding](#), [Zhihu Zhang](#), [Jianwei Gao](#), [Shaletanati Ake](#), [Xuexue Li](#), [Chun Li](#), [Junlei Qi](#), [Jian Cao](#) [Show more](#) ✓

View PDF

[Download full issue](#)[Cite](#)[Add to Mendeley](#)[Share](#)[10.1016/j.cej.2025.169919](#) ↗**Article**[Recommended articles](#)[Metrics](#)

Outline

[Highlights](#)[Abstract](#)[Graphical abstract](#)[Keywords](#)[1. Introduction](#)[2. Experimental](#)[3. Results and discussion](#)

Highlights

- Femtosecond laser multiscale modification enables sintered-joining of BZCYYb and Crofer 22 APU.
- Laser-induced grooves, clusters, and oxygen vacancies synergistically enhance Ag sintering.
- A dense joint with 42.8MPa (97% increase) strength and long-term oxidation stability is achieved.
- Oxygen vacancies cause Jahn–Teller distortion, enhancing d–p hybridization and Ag—O bonding.

Abstract

Reliable ceramic-metal joining is essential for the construction of protonic ceramic fuel cell (PCFC) stacks, yet joint failure still occurs on the electrolyte ceramic side. For the first time, this study proposes a femtosecond laser-enabled multiscale interface engineering strategy to promote the sintered-joining between $\text{BaZr}_{0.4}\text{Ce}_{0.4}\text{Y}_{0.1}\text{Yb}_{0.1}\text{O}_{3-\delta}$ electrolyte and Crofer 22 APU interconnect. The method enables the formation of microscale grooves ($15\mu\text{m}$ width and $2.5\mu\text{m}$ depth), nanoscale clusters (ranging from several tens to hundreds of nanometers), and oxygen vacancies (concentration increased by 100%) at a laser power of 12mW. The resulting hierarchical interface architecture markedly promotes effective sintering of the micro-Ag particle interlayer. Under optimized conditions (930°C , 1 MPa, 30min), the resulting joints are defect-free and exhibit exceptional durability in humid oxidative environments. The shear strength reaches 42.8MPa, representing a 97% improvement over untreated samples. The multiscale enhancement mechanism is systematically elucidated. First-principles calculations further reveal a novel atomic-scale joining mechanism: laser-induced oxygen vacancies trigger localized Jahn-Teller distortion of the crystal field, thereby enhancing the d-p orbital hybridization between the coordinating O and Ag. This femtosecond laser-driven interface engineering strategy offers a robust and scalable solution for high-performance ceramic-metal integration in advanced energy systems.

Graphical abstract

4. Conclusions

CRediT authorship contribution statement

Declaration of competing interest

Acknowledgments

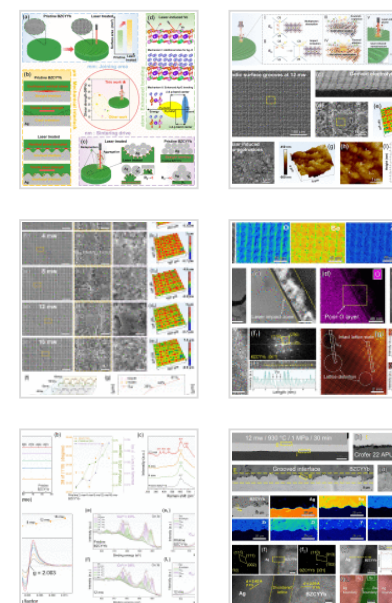
Appendix A. Supplementary data

Data availability

References

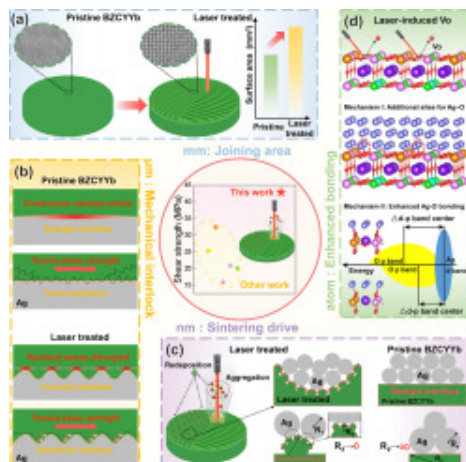
[Show full outline](#) 

Figures (11)



[Show 5 more figures](#) 

Extras (1)



Download: [Download high-res image \(420KB\)](#)

Download: [Download full-size image](#)

Keywords

Protonic ceramic fuel cells; $\text{BaZr}_{0.4}\text{Ce}_{0.4}\text{Y}_{0.1}\text{Yb}_{0.1}\text{O}_{3-\delta}$ electrolyte; Crofer 22 APU interconnect; Micro-Ag interlayer; Femtosecond laser; Multiscale interface regulation

[Previous article in this issue](#)

[Next article in this issue](#)

1. Introduction

In the development of high-efficiency, low-carbon energy systems, proton ceramic fuel cells (PCFCs) have emerged as a promising alternative to conventional solid oxide fuel cells (SOFCs), owing to their lower operating temperatures (400–600°C), higher power densities, and excellent chemical stability [[1], [2], [3]]. Among the various proton-conducting electrolytes, $\text{BaZr}_{0.4}\text{Ce}_{0.4}\text{Y}_{0.1}\text{Yb}_{0.1}\text{O}_{3-\delta}$ (BZCYYb) has attracted considerable attention due to its high proton conductivity, resistance to carbon coking and sulfur

poisoning, and structural stability under dual-atmosphere conditions [4,5]. Consequently, BZCYYb has become a leading electrolyte candidate for next-generation PCFC stacks [6,7]. In practical applications, individual cells must be assembled into stacks to satisfy high power output requirements [8], necessitating reliable, gas-tight joining between the ceramic electrolyte and metallic interconnects, typically ferritic stainless steels such as Crofer 22 APU, because of its CTE compatibility with ceramic electrolytes, formation of a dense and conductive scale with low Cr volatility, and proven brazability and long-term stability [9], [10], [11], [12]].

Compared to conventional glass-based sealants, Ag-based sealing methods have been extensively investigated and successfully applied in SOFCs for joining ceramic electrolytes and metal interconnects, owing to their superior thermal cycling durability and stronger interfacial joining performance [13], [14], [15], [16]]. However, the interface between BZCYYb and stainless steel remains problematic due to mismatches in thermal expansion coefficients and chemical incompatibility [17]. To address these challenges, prior studies have employed a micro-sized Ag particle interlayer to join BZCYYb with stainless steel, utilizing the excellent sinterability and intrinsic plasticity of Ag. This sintered interlayer approach has demonstrated the formation of dense, defect-free joints, while avoiding the formation of brittle interfacial phases commonly observed in Ag-CuO brazing systems [18]. Nevertheless, interfacial failure at the BZCYYb/Ag persists, underscoring the need for strategies that enhance interfacial joining between Ag and BZCYYb.

Femtosecond laser processing, distinguished by its minimal thermal impact, high precision, and contamination-free operation, has emerged as a promising alternative to conventional surface treatments such as mechanical roughening, chemical etching, and plasma processing [19,20]. It leverages nonlinear phenomena such as multiphoton absorption to engineer microgrooves with controlled depth and morphology [21,22]. These microstructures transform planar ceramic interfaces into three-dimensional interlocking architectures, enhancing contact area, redistributing residual stresses, and reinforcing mechanical interlocking at the microscale [23], [24], [25]]. Comparable surface structures could also be realized on BZCYYb ceramics, potentially enlarging the

contact interface and improving interfacial interlocking with Ag, whose plasticity facilitates embedding into grooves during joining [26]. At the nanoscale, femtosecond laser-induced plasma dynamics have been reported to redeposit material, forming nanoparticles and nanoclusters [27,28]. However, the generation, morphology, and distribution of nanoclusters on BZCYYb remain uncharacterized. With high surface energy and large specific surface area, these nanostructures may enhance the sintering kinetics of Ag particles [29]. Moreover, femtosecond laser irradiation is recognized for inducing lattice defects, notably oxygen vacancies [[30], [31], [32]]. Although well-documented in oxide ceramics, such defects in proton-conducting ceramics like BZCYYb remain insufficiently explored. Oxygen vacancies at interfaces are known to strengthen atomic bonding and promote interfacial adhesion [33,34]. Collectively, these insights suggest that femtosecond laser processing offers the potential to enable multiscale interface engineering on BZCYYb ceramics, combining microgrooves, nanoclusters, and atomic-scale defects to facilitate the formation of mechanically robust ceramic/metal joints for high-performance energy systems.

This study introduces and validates a strategy of multiscale interface regulation through femtosecond laser surface grooving, synergistically combined with micro-Ag particle interlayer for sintered-joining BZCYYb and Crofer 22 APU. The femtosecond laser is employed to create a controllable groove structure on the surface of BZCYYb ceramic, enabling the regulation of micro and nanoscale surface morphology and lattice defects. The influence of laser power parameters on interfacial morphology and mechanical properties is systematically investigated. Furthermore, the durability of the resulting joints under humid oxidative environments is evaluated. Computational simulations are employed to elucidate the interfacial stress evolution and joining mechanisms. The findings offer a theoretical framework and practical pathway for the development of high-performance and aging-resistant PCFC power stacks.

2. Experimental

2.1. Materials synthesis

BZCYYb was prepared using the solid-state reactive sintering (SSRS) method, and the details of the preparation process are available in our previous work [35]. The prepared BZCYYb electrolyte exhibits a single perovskite structure, and the position of its XRD diffraction peak aligns with findings reported in Fig. S1 [5]. Crofer 22 APU stainless steel (3mm thick) was employed as the interconnect material, purchased from VDM Metals, Werdohl, Germany. The chemical composition of the steel is 22.9 Cr, 0.44 Mn, 0.1 La, 0.06 Ti, 0.01 Si, 0.01 Al, and balance Fe (in wt%). The steel was cut into specimens with a size of 10mm×10mm×3mm by an electrical discharge machining. The surface to be sealed was polished using 400#, 800#, 1000# and 1500# sandpapers. Ag powder (Aladdin, Shanghai, China 99.98%, ≤5μm) was pressed (200MPa, 180s) into a ~300μm-thick tablet via a tableting instrument. Prior to joining, all materials were ultrasonic cleaned in ethanol for 20min.

2.2. Femtosecond laser surface grooving

The surface of the BZCYYb ceramic was modified using a femtosecond laser (wavelength: 1030nm, pulse width: 300fs, repetition rate: 100kHz) to introduce hierarchical grooves. A cross-patterned structure was designed using a galvanometric scanning system. The laser fluence and scanning parameters were adjusted to produce grooves with varying depths and periodicities. Laser powers of 2–16mW were applied to study the effects of processing energy on surface morphology. The scanning speed was fixed at 3m/min with a hatch spacing of 20μm.

2.3. Joining process

A micro-Ag particle interlayer was used as the diffusion medium and sandwiched between the laser-grooved surface of the BZCYYb ceramic and the polished Crofer 22 APU stainless steel. The assembly was aligned and stacked into a layered configuration: BZCYYb/Ag/Crofer 22 APU, as shown in the supplementary material (Fig.S2a). Sintered joining was performed in a tubefurnace under atmospheric conditions. The specimens

were heated to 930°C at a rate of 3°C/min, held at peak temperature for 30 min under a uniaxial pressure of 1 MPa, and then cooled to room temperature at the same rate (3°C/min).

2.4. Long-term test in oxidizing atmospheres

The obtained joints were placed in a horizontal tube furnace (OTF-1200×) and thermally aged at 600°C for 300 h (3°C/min heating/cooling) in humidified air (50% air/50% H₂O; 6 Lh⁻¹) to simulate the cathode-side oxidizing and hydrated environment typical of PCFC operation [[1], [2], [3], [4], [5]].

2.5. Microstructural characterization

The surface and interfacial morphologies were characterized using scanning electron microscopy (SEM, Zeiss Merlin) and focused ion beam (FIB, FEI NOVA 400). The chemical composition and element distributions were analyzed by energy-dispersive X-ray spectroscopy (EDS) and wavelength-dispersive spectroscopy (WDS). X-ray diffraction (XRD, PANalytical Empyrean, Cu K α) and Raman spectroscopy (Renishaw inVia) were used to analyze crystal structure and defect evolution. Atomic force microscopy (AFM, Bruker Dimension Icon) and 3D white light interferometry were used for topographical mapping, with the latter providing quantitative data such as groove depth, width, and surface area growth rate. High-resolution transmission electron microscopy (HRTEM, Talos F200X) was conducted to investigate atomic-scale lattice changes and interfacial features. Electron paramagnetic resonance (EPR, Bruker EMX) and X-ray photoelectron spectroscopy (XPS, PHI 5000 VersaProbe III) were used to quantify oxygen vacancy concentration. The shear tests are carried out at room temperature via the electronic universal testing machine (Instron 1186, Britain). The schematic illustration of the shear test is displayed in Fig. S2b. Five samples were averaged to get the shear strength values at each parameter.

2.6. Shear test and ASR measurement

The shear tests are carried out at room temperature via the electronic universal testing machine (Instron 1186, Britain). The schematic illustration of the shear test is displayed in Fig. S2b. Five samples were averaged to get the shear strength values at each parameter. The area-specific resistance (ASR) of the BZCYYb/Ag/Crofer 22 APU assembly was measured using the four-probe method. The assembled stack was placed in a custom-designed tubular furnace and heated to 900°C at a rate of 5°C/min, followed by cooling to 500°C at the same rate. During the cooling process, voltage drops were recorded at 20°C intervals, and the ASR at each temperature was calculated accordingly.

2.7. DFT calculation

All calculations were performed using spin-polarized density functional theory (DFT) within the projector augmented wave (PAW) method, as implemented in VASP [36]. The exchange-correlation potential was treated using the generalized gradient approximation (GGA) by Perdew, Burke, and Ernzerhof (PBE) [37], and van der Waals interactions were accounted for using the DFT-D3 scheme [38]. The DFT+U approach was applied with effective U values of 2.08 eV for Y, 2.68 eV for Zr and 5.00 eV for Ce to correct d electron correlation effects [39,40]. A plane-wave cutoff energy of 480 eV and a convergence criterion of 10^{-5} eV for electronic iterations were adopted. Structural relaxations continued until the residual atomic forces were below 0.02 eV/Å. The Brillouin zone was sampled using a $4 \times 1 \times 1$ k-point mesh. The Ag(111)/BZCYYb(110) heterojunction was constructed based on HRTEM results, exhibiting lattice mismatches of 2.54% (a-direction) and 3.64% (b-direction). The model includes 64 Ag, 15 Ba, 6 Ce, 2 Y, 6 Zr, 1 Yb, and 45 O atoms. Valence electrons are mainly distributed in the Ag (5s, 4d), Ba (6s), Ce (4f, 5d, 6s), Y (4d, 5s), O (2s, 2p), Zr (4d, 5s), and Yb (4f, 6s) orbitals. An oxygen vacancy was introduced by removing one O atom. To eliminate inter-slab interactions, a vacuum layer of 20 Å was added. Details of the model construction are provided in the Supporting Information (Fig. S3).

The work of adhesion is defined as the energy (per unit area) needed to reversibly separate an interface into two free surfaces, which can quantitatively evaluate the

bonding strength of the resulting interfaces. The work of adhesion can be calculated by [41]:

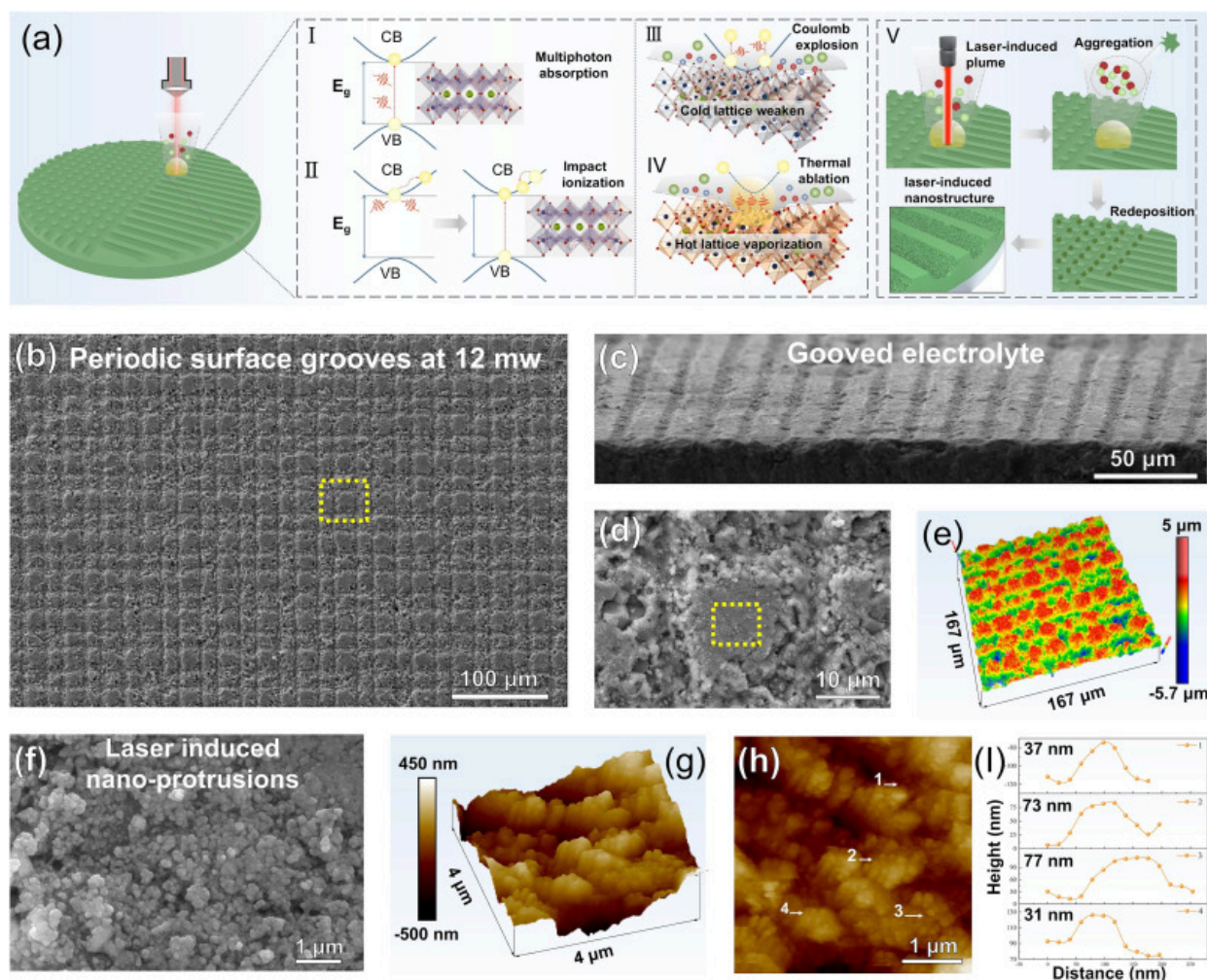
$$W_{adh} = (E_{Ag} + E_{BZCYYb} - E_{Ag/BZCYYb}) / A \quad (1)$$

Where $E_{Ag/BZCYYb}$ the total energy of the Ag (111)/BZCYYb (110) interface. E_{Ag} and E_{BZCYYb} represent the energies of the Ag (111) slab and BZCYYb (110) slab after structural relaxation, respectively. In addition, A is the area of the interface model. We define $\Delta\rho = \rho_{A+B} - \rho_A - \rho_B$ as the charge density difference of A/B heterostructure, where ρ_{A+B} , ρ_A and ρ_B are the charge densities of A/B heterostructure, isolated A and B slabs, respectively. We use the Bader charge to express the charge transfer quantity. COHP and ICOHP for selected supercells are calculated using the LOBSTER programme.

3. Results and discussion

3.1. Evolution of surface structure and lattice defect on laser-modified BZCYYb ceramic

Femtosecond laser grooving on BZCYYb ceramics involves ultrafast electronic excitation, material removal, and hierarchical structure formation [42,43]. As shown in Fig. 1a, multiphoton absorption initiates carrier excitation due to the wide bandgap ($\sim 2\text{eV}$) [44], followed by impact ionization that rapidly increases free electron density (Fig. 1aI-II). Once the critical threshold is reached, Coulomb explosion disrupts the lattice nonthermally (Fig. 1aIII), while electron-phonon coupling induces thermal ablation (Fig. 1aIV). The resulting plasma plume aggregates and redeposits particles, leading to the formation of nanostructures (Fig. 1aV).



Download: [Download high-res image \(2MB\)](#)

Download: [Download full-size image](#)

Fig. 1. Formation mechanism and characteristic morphology of femtosecond laser-induced surface patterning. (a) Schematic representation of the multi-physics interactions governing femtosecond laser-material processing and nanoparticle formation. (b) Microstructural of the grooved pattern. (c) Cross-sectional morphology of the laser-etched grooves. (d) Magnified microstructure of the laser-processed region in Fig. 2b. (e) 3D topographic of the grooved pattern. (f) Enlarged view of the femtosecond

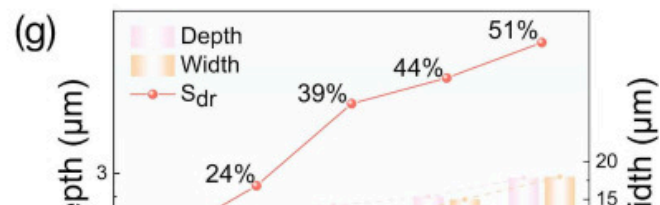
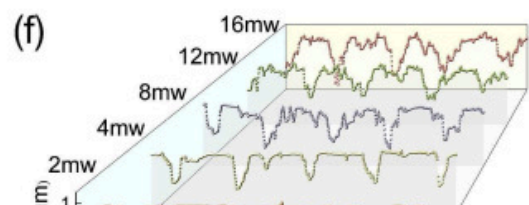
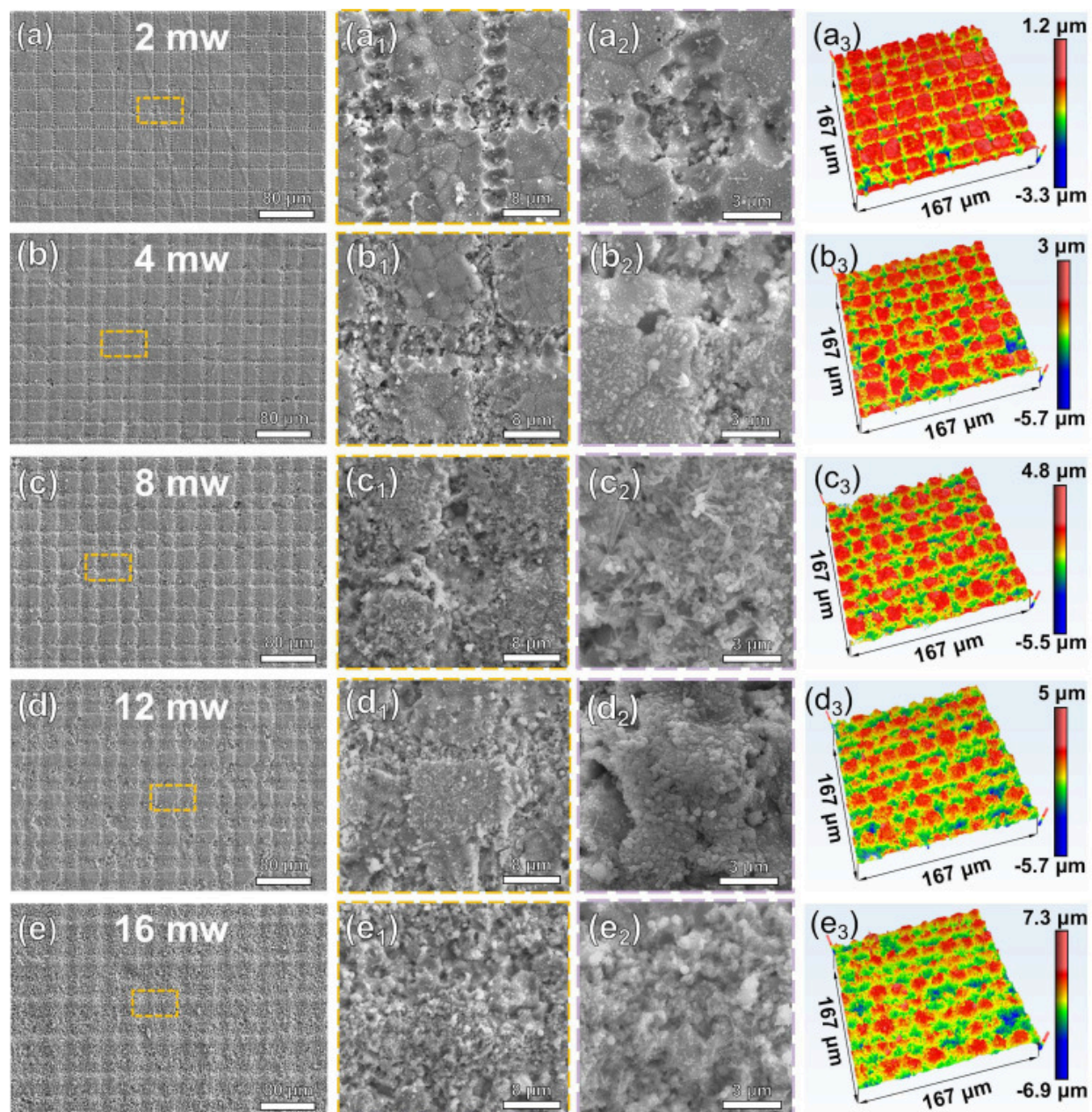
laser-induced nanoscale particle formation in Fig. 2d. (g) 3D topographic of the laser-induced nanostructures. (h) Microscopic morphology of laser-induced nanostructures. (i) The height of nanoparticles.

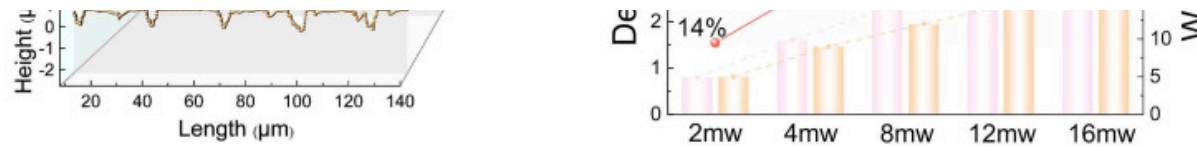
Fig. 1b demonstrates the successful fabrication of a large-scale, uniform, and repeatable cross-patterned groove structure on the ceramic surface by femtosecond laser. Cross-sectional analysis in Fig. 1c reveals that the grooves maintain a consistent width of approximately $15\mu\text{m}$, exhibit uniform depth, and possess well-defined bottom boundaries without melt accumulation. This indicates that the ultrashort pulse duration of the femtosecond laser effectively suppresses heat diffusion, thereby confining the heat-affected zone to a localized region [45]. Fig. 1d magnifies the area marked by the yellow box in Fig. 1b, revealing micro and nanoscale features along the groove sidewalls and bottom. To further quantify surface topographical changes, Fig. 1e presents a white-light interferometry map of the laser-grooved surface over a $167\mu\text{m} \times 167\mu\text{m}$ area, with height variations from $-5.7\mu\text{m}$ to $5\mu\text{m}$. The grooves are measured to be $15\mu\text{m}$ wide and approximately $2.5\mu\text{m}$ deep, corresponding to a 44% increase in surface area compared to the untreated surface.

The formation of micro and nanostructures are further confirmed at finer scales. Fig. 1f provides a magnified view of the yellow-boxed area in Fig. 1d, illustrating the emergence of nanoscale protrusions on the ceramic surface following the laser treatment. These protrusions, ranging from tens to hundreds of nanometers, exhibit irregular morphologies and form aggregated nanoclusters in certain regions. Further atomic force microscopy (AFM) analysis, shown in Fig. 1g-1i, reveals the 3D morphology of the microregion over a $4\mu\text{m} \times 4\mu\text{m}$ area. The surface heights is varied from -500nm to 450nm . The surfaces of micro-protrusions are covered with nanoparticles and nanoclusters, typically tens of nanometers in height (Fig. 1i), forming a hierarchical micro and nanostructure with multiscale characteristics.

Fig. 2 illustrates the evolution of surface groove morphology and laser-induced nanostructures under various femtosecond laser powers. Fig. 2a-2e show the macroscopic

changes in groove structures, while Fig. 2a₁–2e₁ provide magnified views of the laser-processed regions. The grooves exhibit blurred boundaries and incomplete etching at low power levels of 2mW (Fig. 3a) and 4mW (Fig. 2b). Enlarged views (Fig. 2a₁–2b₁) reveal periodic microcraters due to the Gaussian beam profile, where only the central region exceeds the ablation threshold, resulting in discrete cratering rather than continuous grooves. In contrast, the grooves exhibit well-defined edges and distinct periodicity at 8mW and 12mW (Fig. 2c–2d), indicating that the combined effects of non-thermal ablation and localized thermal melting contribute to stable material removal. However, the groove boundaries become less distinct at 16mW (Fig. 2e₁), suggesting overlapping energy deposition degrades etching precision.

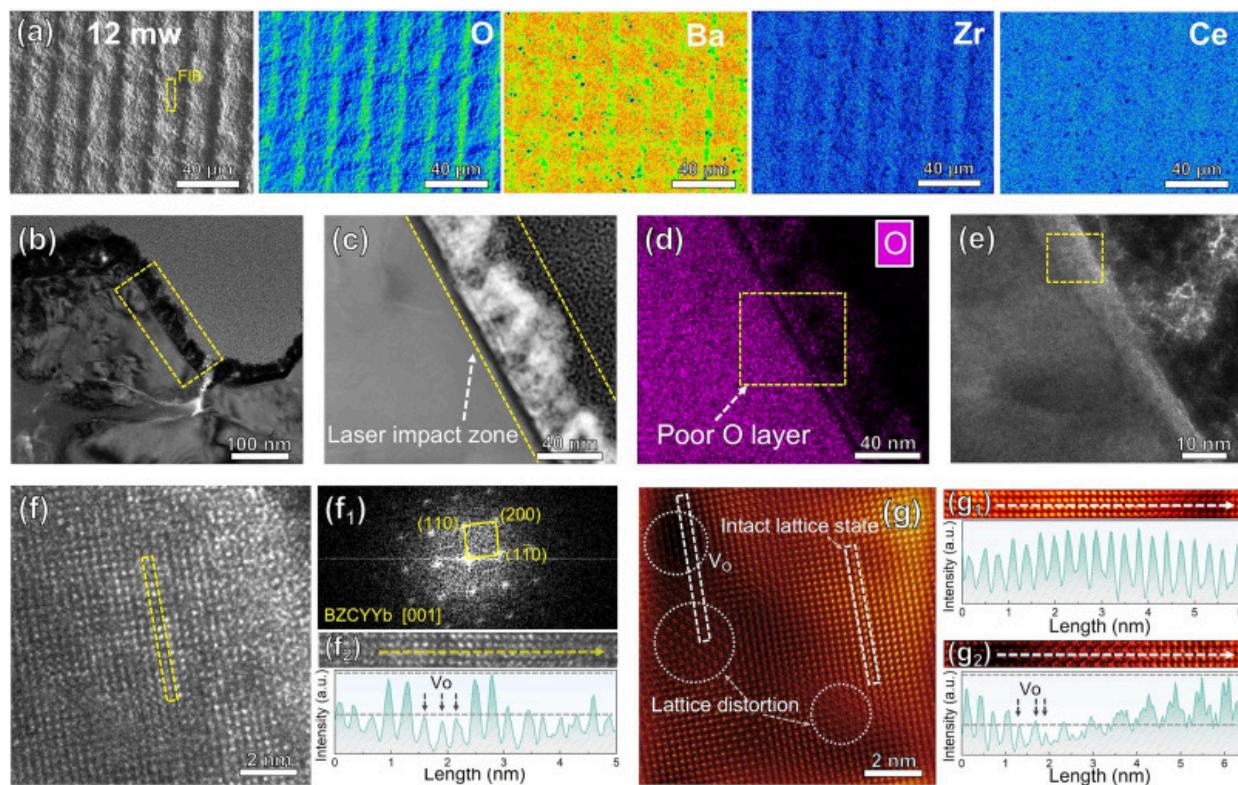




[Download: Download high-res image \(4MB\)](#)

[Download: Download full-size image](#)

Fig. 2. Surface groove morphology and micro/nanostructure evolution on BZCYYB ceramic under different femtosecond laser powers. (a-e) Microscopic morphology of surface grooves at various laser powers. (a₁-e₁) Enlarged views of the corresponding groove microstructures from (a-e). (a₂-e₂) evolution of surface nanomicro-structures with increasing laser power. (a₃-e₃) the 3D topographic of groove morphology at various laser powers. (f) Vertical cross-sectional profiles of grooves at varying laser powers. (g) Quantitative analysis of groove depth, width, and surface area growth rate at various laser powers.



[Download: Download high-res image \(3MB\)](#)

[Download: Download full-size image](#)

Fig. 3. Characterization of femtosecond laser-induced oxygen vacancies in BZCYYb ceramics. (a) Surface morphology and elemental distribution after laser grooving; (b) Bright-field TEM (BF-TEM) image; (c) Magnified HAADF-STEM image of (b); (d) EDS mapping of oxygen; (e) Magnified HRTEM image; (f) Further magnified HRTEM image of (e); (f₁) FFT pattern of (f); (f₂) Atomic intensity profile of (f); (g) IFFT image; (g₁) Atomic intensity profile of intact lattice in (g); (g₂) Atomic intensity profile of oxygen vacancy layer in (g).

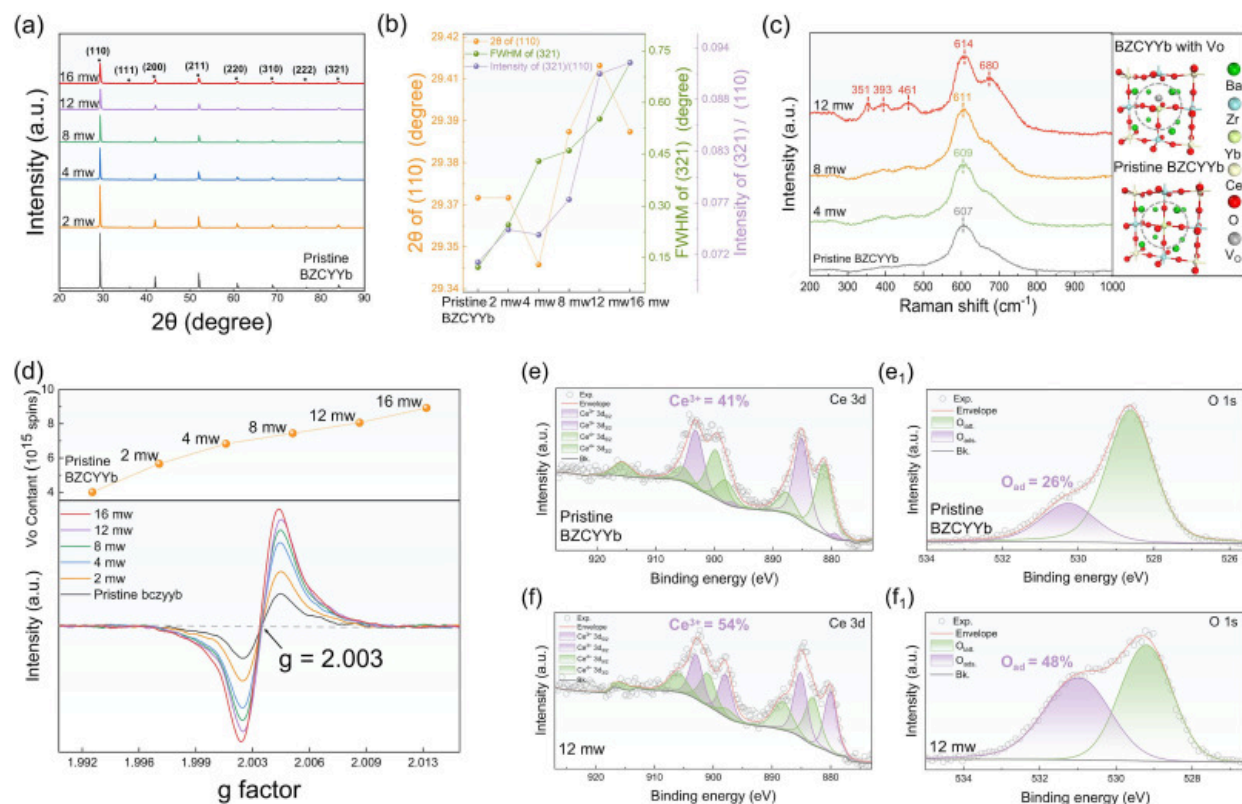
Fig. 2a₂-2e₂ illustrates the morphology and distribution of femtosecond laser-induced nanostructures. Sparse nanoparticles are uniformly distributed across the surface at 2–4 mw (Fig. 2a₂-2b₂). As laser power increases, the nanoparticle coverage is intensified,

accompanied by distinct morphological transformations. Nanoparticles appear in point-like and rod-like forms at 8mW (Fig. 2c₂). Further increasing the laser power to 12mW, they aggregate into spherical clusters (Fig. 2d₂), suggesting that plasma plume condensation and material redeposition become more pronounced, promoting the formation of high-energy nanoclusters that may enhance sintering during subsequent joining because of their extremely small curvature radius and the resulting high surface energy rather than chemical composition [29]. However, nanoparticle density decreases significantly at 16mW (Fig. 2e₂), indicating that excessive laser energy may induce secondary ablation [46].

Fig. 2a₃-2e₃ present white-light interferometry results of the 3D groove morphology. Analysis of the 167μm×167μm region reveals that surface undulations increase with rising laser power. Fig. 2f shows the depth profiles extracted from the 3D morphology data, indicating that from 2mW to 12mW, the grooves exhibit periodic and well-defined depth variations, suggesting a stable and controlled ablation process. However, the periodicity deteriorates and the profile becomes irregular at 16mw, reflecting inhomogeneous material removal and reduced etching accuracy. Fig. 2g quantifies the growth of groove depth, width, and the surface area growth rate across different laser power levels. Groove depth increases from 0.8μm at 2mW to 2.9μm at 16mW, while width expands from 5μm to 18μm. Correspondingly, the surface area growth rate rises from 14% to 51%. The growth rate is defined as the percentage increase of the actual surface area relative to the projected area, and these values are automatically quantified by the instrument software.

To investigate the effects of femtosecond laser irradiation on BZCYYb ceramics, Fig. 3a shows periodic grooves formed at a laser power of 12mW, along with elemental maps of O, Ba, Zr, and Ce. Due to the relatively low evaporation threshold of Ba, its content slightly decreases within the laser-processed region [47], although the BZCYYb phase remains stable, as confirmed in Fig. 3f. The oxygen content is noticeably reduced in the laser-affected zone, while the distributions of Ce, Zr, Y, and Yb remain largely unchanged. The distributions of Y and Yb are shown in the Supporting Information (Fig. S4), and for

comparison, SEM images and elemental maps of the pristine ceramic surface are provided in Fig. S5. Fig. 3b presents a bright-field transmission electron microscopy (TEM) image of the laser-affected region, revealing a nanoparticle-rich layer composed of Y- and Yb-doped ZrO, which forms as a result of preferential Ba evaporation [48]. The elemental composition of the nanoparticles is detailed in the Supplementary Information (Table S1). The high-angle annular dark field (HAADF) image (Fig. 3c) shows a transition layer between the nanoparticle-rich region and the substrate. The corresponding energy-dispersive spectroscopy (EDS) mapping (Fig. 3d) reveals oxygen depletion within this layer, which may be associated with the formation of oxygen vacancies. Quantitative analysis of oxygen vacancies concentration following laser treatment is presented in the subsequent section (Fig. 4).



[Download: Download high-res image \(1MB\)](#)

[Download: Download full-size image](#)

Fig. 4. Quantitative study of femtosecond laser-induced oxygen vacancies on the surface of BZCYYb ceramics. (a) XRD diffraction patterns under different laser power treatments; (b) XRD quantification (2θ angle, FWHM and peak intensity ratio); (c) Raman spectra and atomic structure schematic at different laser powers; (d) EPR spectra and oxygen vacancy concentration; (e-e₁) Ce 3d and O 1s XPS spectra in the pristine state; (f-f₁) Ce 3d and O 1s XPS spectra after 12mW laser treatment.

Atomic-scale characterization is presented in the HRTEM image in Fig. 3e, with magnified views in Figs. 3f-3g. Fig. 3f reveals partial amorphization near the nanoparticles, implying laser-induced remelting and possible phase transitions [49]. The corresponding FFT pattern (Fig. 3f₁) exhibits diffraction spots corresponding to the (110) and (200) planes, confirming the preservation of the BZCYYb phase. The atomic column profile extracted in Fig. 3f₂ displays non-uniform contrast along the column, which likely arises from lattice distortion induced by oxygen vacancy clustering [50]. The IFFT image in Fig. 3g more clearly depicts structural inhomogeneities caused by oxygen vacancy-related B—O bond disruptions and octahedral distortions. Fig. 3g₂ quantifies these effects via intensity fluctuation and peak splitting in oxygen vacancy-rich regions, suggesting localized lattice strain. In contrast, Fig. 3g₁ shows uniformly ordered lattice columns in intact regions. These stark contrasts provide direct evidence of oxygen vacancy-induced lattice deformation following femtosecond laser exposure [51].

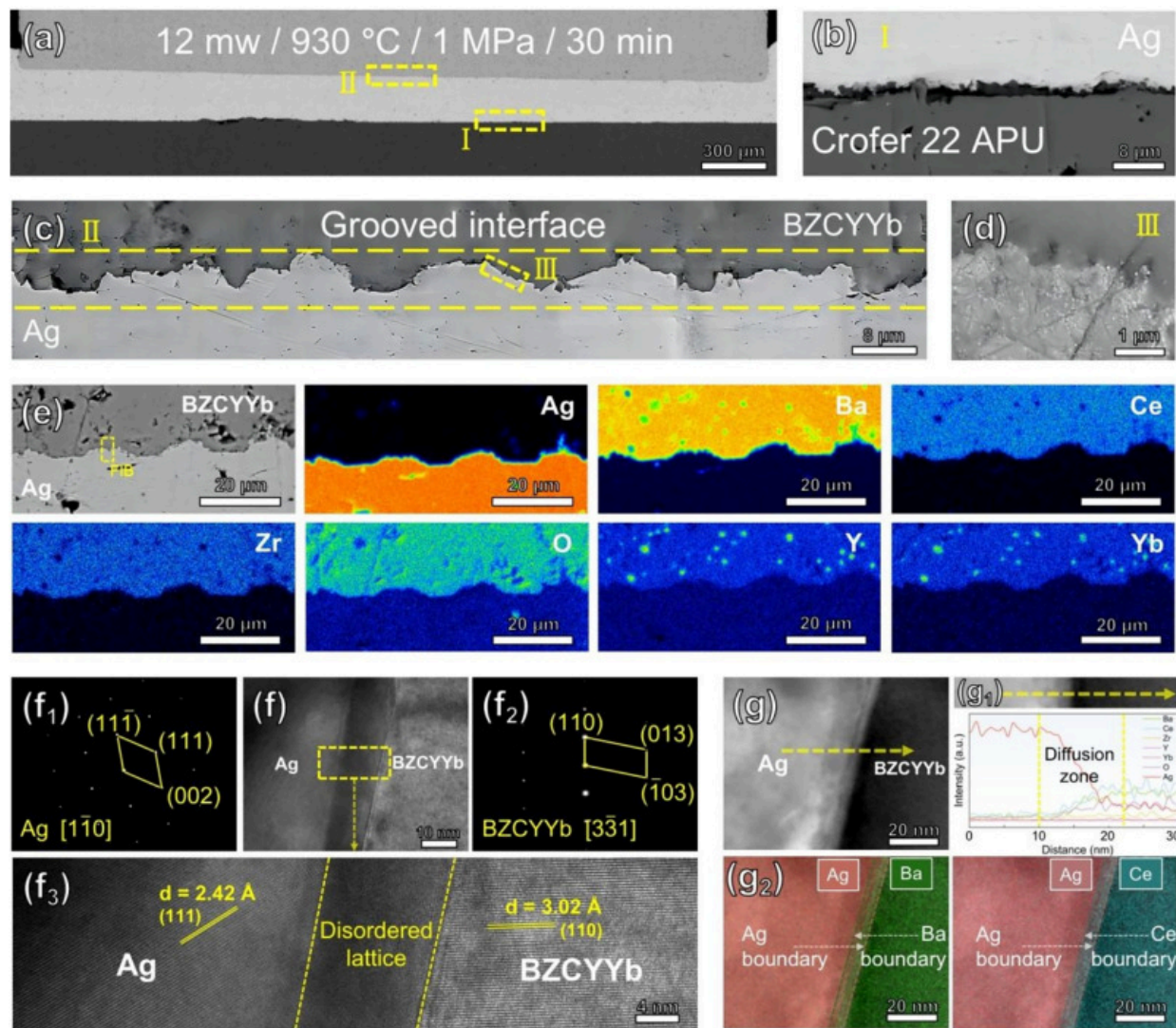
Fig. 4a presents the X-ray diffraction (XRD) patterns of BZCYYb ceramics processed under different femtosecond laser powers (2–16mW). The absence of new diffraction peaks confirms that no detectable phase transformation occurs in the BZCYYb ceramic matrix after laser treatment. The Y- and Yb-doped ZrO nanoparticles reside within a shallow surface region (a few tens of nanometers in depth), which is beyond the detection capability of XRD. However, Fig. 4b shows a non-monotonic shift in the 2θ position of the (211) plane, reflecting the competing effects between residual compressive stress and oxygen vacancy formation. Compressive stress shifts 2θ to higher angles, while oxygen vacancy expansion reduces electrostatic shielding around B-site cations (Zr, Ce, Yb, Y), increasing B—O bond lengths and expanding the lattice [52]. The increasing full-width at

half maximum (FWHM) of the (321) peak confirms laser-induced lattice strain, while the rising (321)/(110) peak intensity ratio suggests greater exposure of high-index planes, potentially enhancing sintering behavior. Fig. 4c further illustrates laser-induced structural modifications through Raman spectroscopy. New vibrational modes at 351, 393, 461, and 680 cm^{-1} emerge and intensify with increasing laser power. The low-frequency peaks correspond to asymmetric B—O stretching and bridge-bond vibrations caused by oxygen vacancies, while the high-frequency peak reflects internal vibrations within the $[\text{BO}_6]$ octahedra, indicating local coordination rearrangement [53]. Moreover, the primary Raman peak shifts from 607 cm^{-1} to 614 cm^{-1} due to residual compressive stress. The adjacent atomic models highlight that oxygen vacancy disrupts local B—O bonds, leading to bond bending and octahedral Jahn-Teller distortion [54].

To quantify oxygen vacancy concentration, electron paramagnetic resonance (EPR) and X-ray photoelectron spectroscopy (XPS) were conducted. Fig. 4d shows EPR spectra featuring a Lorentzian signal at $g=2.003$, characteristic of unpaired electrons trapped at OV sites [55]. Signal intensity rises with laser power, peaking at 16mW, with vacancy density reaching 8.9×10^{15} spins, more than twice that of the pristine sample (4×10^{15} spins). This trend aligns with XRD and Raman results. XPS further supports these findings: Fig. 4f-4g show Ce 3d and O 1s spectra. The Ce^{3+} content increases from 41% (pristine) to 54% at 12mW, confirming $\text{Ce}^{4+} \rightarrow \text{Ce}^{3+}$ reduction induced by oxygen vacancies formation. Since each oxygen vacancy reduces two Ce^{4+} ions, Ce^{3+} serves as an indirect indicator of oxygen vacancy concentration [56,57]. The O 1s spectra further validate this trend. Deconvolution reveals two components at $\sim 529.5\text{eV}$ (lattice oxygen, O_{latt}), $\sim 531\text{eV}$ (adsorbed oxygen, O_{ads}) [58]. The O_{ads} peak, associated with oxygen species adsorbed at vacancies, increases from 26% (pristine) to 48% (12mW-treated), demonstrating a significant rise in surface oxygen vacancies. In summary, the effective induction of oxygen vacancies by femtosecond laser processing has been conclusively demonstrated. Subsequent studies confirm that oxygen vacancies facilitates the sintering of Ag particles at interface and enhances interfacial bonding strength (Fig. 9).

3.2. Interfacial microstructure of laser-grooved BZCYYb/Crofer 22 APU joints

As previously mentioned, Ag-based materials have been widely applied in solid oxide cell packaging due to their excellent thermal cycling durability and interfacial compatibility with both ceramic electrolytes and metallic interconnects [[13], [14], [15], [16]]. In this study, a micro-Ag particle interlayer (average particle size $\sim 5\mu\text{m}$) is employed to achieve sintered-joining between the electrolyte ceramic and Crofer 22 APU. The influence of surface micro and nanostructures, formed under various femtosecond laser powers, on the interfacial microstructure of the joints is also investigated. Fig. 5a illustrates the cross-sectional microstructure of the joint formed at 930°C for 30min under 1 MPa pressure after processing at 12mW. The BZCYYb electrolyte (upper layer) is tightly joined to the Crofer 22 APU through an Ag particle interlayer. A magnified view of the stainless steel-side interface (Fig. 5b) reveals a uniform oxide layer approximately $2\mu\text{m}$ thick, primarily composed of $(\text{Cr,Fe,Ag})_2\text{O}_3$ (Table S2), consistent with previous reports [59]. Compared to conventional Ag-CuO brazed joints, which typically exhibit oxide layer exceeding $20\mu\text{m}$ [60,61], the significantly thinner layer is advantageous for ensuring reliable sealing and maintaining long-term joint stability [62]. Fig. 5c presents the microscopic morphology of the BZCYYb/Ag interface, showing a well-defined groove structure fully infiltrated by Ag, forming a compact joining. The absence of an interfacial reaction layer in the SEM magnification image (Fig. 5d) confirms that the joining process is primarily diffusion-driven rather than chemically induced. Elemental diffusion behavior is characterized by WDS (Fig. 5e), showing a nanoscale Ag diffusion zone tapering into BZCYYb. Ba displays a broader diffusion profile, likely due to its higher reactivity. In contrast, Ce, Zr, and O exhibit no significant segregation, confirming the absence of brittle reaction phase. The diffusion zones of Y and Yb could not be clearly distinguished in the WDS analysis, due to their relatively low concentrations (10% of the total B-site).



[Download: Download high-res image \(839KB\)](#)

[Download: Download full-size image](#)

Fig. 5. Microstructure of the BZCYYb/Ag/Crofer 22 APU joint obtained at 930°C, 1 MPa, for 30min with 12mW femtosecond laser grooving: (a) The interface of the BZCYYb/Ag/Crofer 22 APU joint; (b) Magnified view of the Ag interlayer/Crofer 22 APU interface; (c) Enlarged view of the laser-grooved BZCYYb/Ag interface; (d) High-

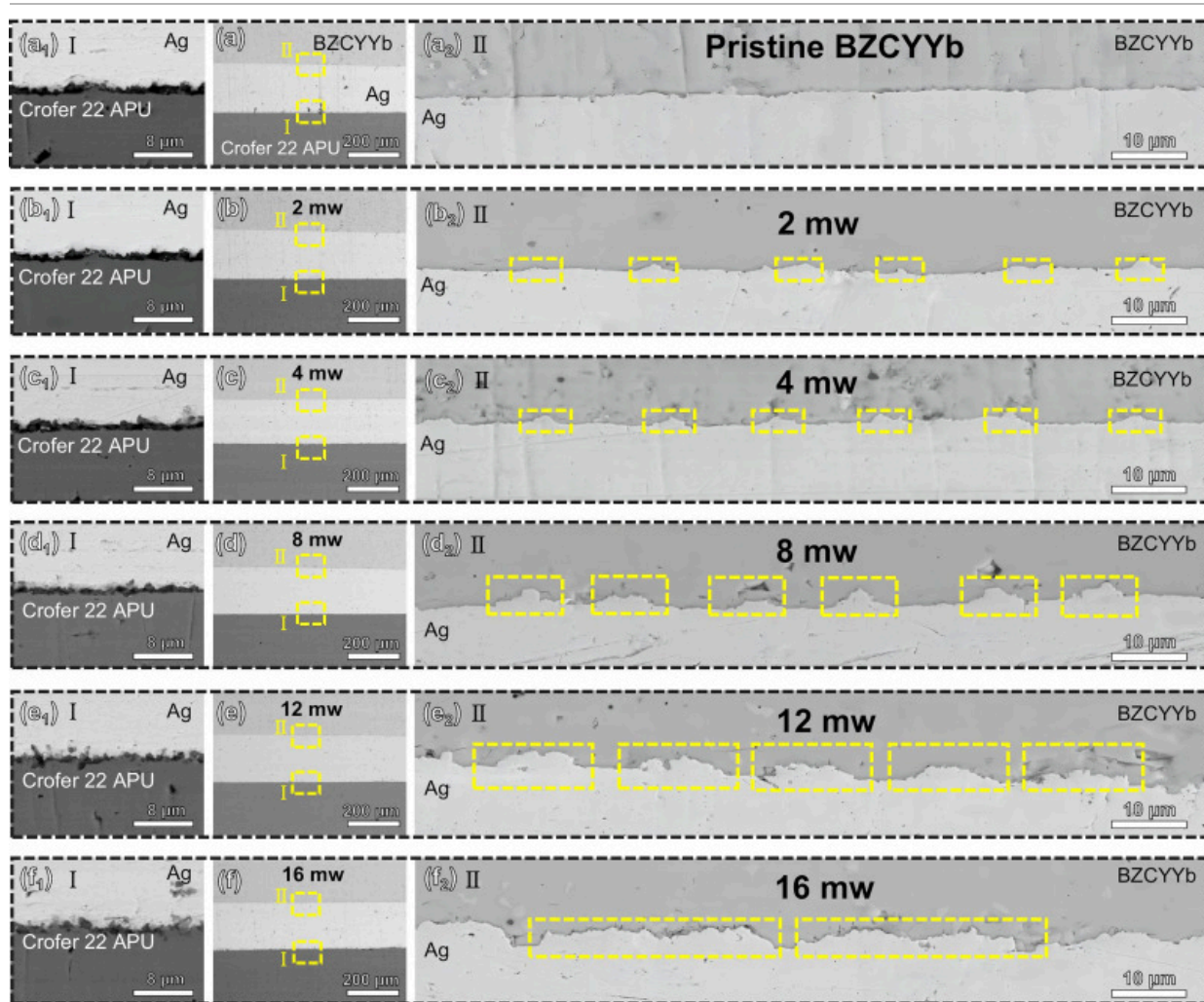
magnification SEM image of the BZCYYb/Ag interface; (e) Elemental distribution at the BZCYYb/Ag interface; (f) HRTEM analysis of the BZCYYb/Ag interface; (f₁) SAED pattern of Ag; (f₂) SAED pattern of BZCYYb; (f₃) Enlarged HRTEM view of (f); (g) STEM image of the BZCYYb/Ag interface; (g₁) EDS line scan across the interface; (g₂) Elemental mapping of the interface.

To elucidate the joining mechanism, TEM analysis was conducted (Fig. 5f). HRTEM reveals a clear Ag/BZCYYb interface, confirming interatomic-level joining. A magnified region (Fig. 5f₃) shows three distinct zones: crystalline Ag (111) planes ($d=0.242\text{ nm}$, SAED in Fig. 5f₁), crystalline BZCYYb (110) planes ($d=0.302\text{ nm}$, SAED in Fig. 5f₂), and a $\sim 10\text{ nm}$ thick disordered interfacial layer, attributed to interdiffusion. This region is critical for interatomic joining, a phenomenon reported in other diffusion joining studies [63]. Moreover, the Y- and Yb-doped ZrO nanoparticles induced by Ba evaporation during laser processing are absent at the interface, most likely because Ba diffuses from the bulk under thermal and pressure-driven conditions, replenishing the Ba-deficient regions and thereby reforming the BZCYYb phase, which is consistent with the observations of Chen et al. [64], rather than being dissolved or buried by Ag infiltration. The STEM image (Fig. 5g) clearly distinguishes the BZCYYb phase (dark contrast) from the Ag phase (bright contrast). EDS analysis along the designated yellow line (Fig. 5g₁) identifies a $\sim 12\text{ nm}$ diffusion zone marked by decreasing Ag and increasing Ba/Ce signals. The signals of Y and Yb fall below the reliable detection threshold, owing to their low concentrations. The elemental distribution map (Fig. 5g₂) further confirms Ag—Ba and Ag—Ce interdiffusion, with a diffusion width of approximately 10 nm , consistent with the EDS line scan results, suggesting that interdiffusion contributes to the formation of the disordered lattice region observed in Fig. 5f₃.

3.3. Effect of laser power on microstructure of BZCYYb/Crofer 22 APU joints

Fig. 6a–6f presents the interfacial microstructures of joints obtained using an Ag particle interlayer at various femtosecond laser powers (2mW, 4mW, 8mW, 12mW, and 16mW). Apparently, the BZCYYb electrolyte and Crofer 22 APU are soundly joined by the Ag

interlayer. The magnified view at the Ag/Crofer 22 APU interface (Fig. 6a₁-6f₁) reveals a uniform and dense oxide layer ($\sim 2\mu\text{m}$ thick) across all laser power conditions. Fig. 6a₂-6f₂ highlights the morphological evolution of the BZCYYb/Ag interface under various laser powers. The Ag interlayer effectively embeds into the grooves, ensuring intimate contact with the BZCYYb electrolyte. In the ungrooved condition (Fig. 6a₂), the interface appears relatively flat. Shallow grooves emerge with minor deviation from the unprocessed surface at 2mW (Fig. 6b₂). As the laser power increases to 4mW (Fig. 6c₂), a periodic groove pattern emerges, albeit with limited depth. Both groove depth and width increase significantly at 8mW (Fig. 6d₂), leading to a more pronounced structure. However, from 2mW to 8mW, the grooves retain sharp bottoms, potentially causing localized stress concentrations. When the laser power reaches 12mW (Fig. 6e₂), the groove depth and width further increase, while the groove bottoms become smoother. This transition results in a well-defined periodic undulation, optimizing interfacial stress distribution and increasing the effective joining area. However, adjacent grooves merge at 16mW (Fig. 6f₂), disrupting the periodic structure and reducing interfacial uniformity. This irregular morphology may compromise joint strength and negatively affect long-term joint stability [21,22].



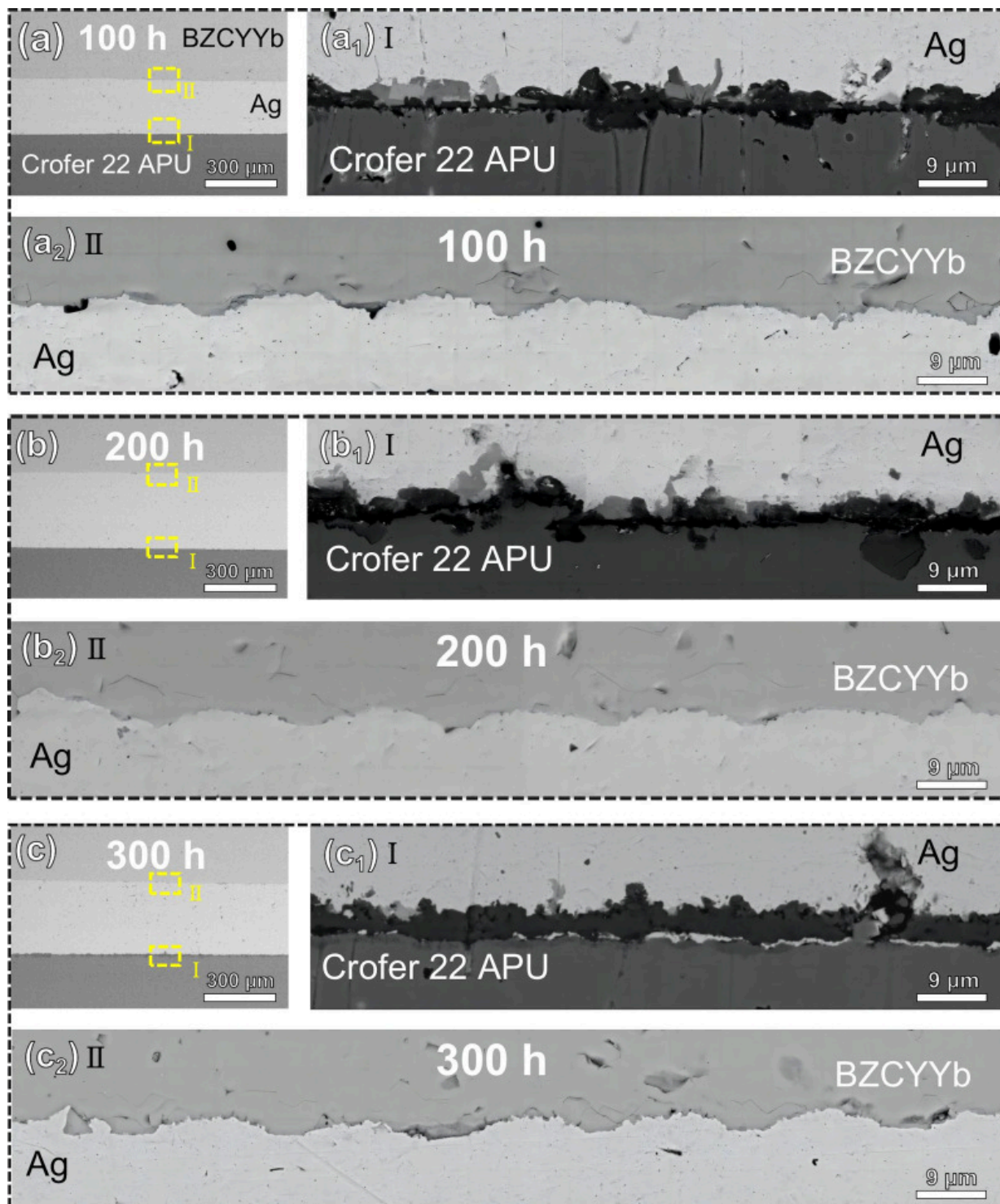
[Download: Download high-res image \(2MB\)](#)

[Download: Download full-size image](#)

Fig. 6. Effect of various power laser grooving on the joint microstructure at 930°C/1 MPa/30 min: (a-f) Microstructure of BZCYYb/Ag/Crofer 22 APU joints with various powers of femtosecond laser grooving; (a_1 - f_1) Enlarged images of Ag/Crofer 22 APU interface; (a_2 - f_2) Microstructure evolution of BZCYYb/Ag interface.

3.4. Stability of the joints in oxidizing atmosphere

To evaluate the long-term stability of the joints in a humid oxidizing atmosphere, aging tests were conducted at 600°C in a 50% air-50% H₂O environment for 100, 200, and 300h, followed by microstructural analysis. Fig. 7a shows that the joint remains dense and defect-free after 100h. As depicted in Fig. 7a₁, the oxide layer along the stainless steel interface remains continuous and compact, without significant thickness change. It primarily consists of (Cr_{0.5}Ag_{0.5})₂O₃ (Table S3) near the Ag interlayer (gray region) and (Cr,Fe,Ag)₂O₃ near the stainless steel (black region). Fig. 7a₂ confirms that the BZCYYb/Ag interface shows no structural degradation. After aging for 200h, the joint maintains structural integrity without defects, as shown in Fig. 7b. The oxide layer at the stainless steel interface slightly thickens but remains uniform and dense (Fig. 7b₁). Similarly, the BZCYYb/Ag interface exhibits no microstructural changes (Fig. 7b₂). The joint remains structurally intact and free of defects after aging for 300h (Fig. 7c). The oxide layer at the stainless steel interface undergoes minor thickening (~5μm) but remains continuous and dense, primarily composed of (Cr,Fe,Ag)₂O₃ (Fig. 7c₁). Meanwhile, the BZCYYb/Ag interface shows no observable degradation (Fig. 7c₂), reinforcing the stability of the diffusion-driven interfacial joining in an oxidative atmosphere and fully validating the long-term stability of this joining method.



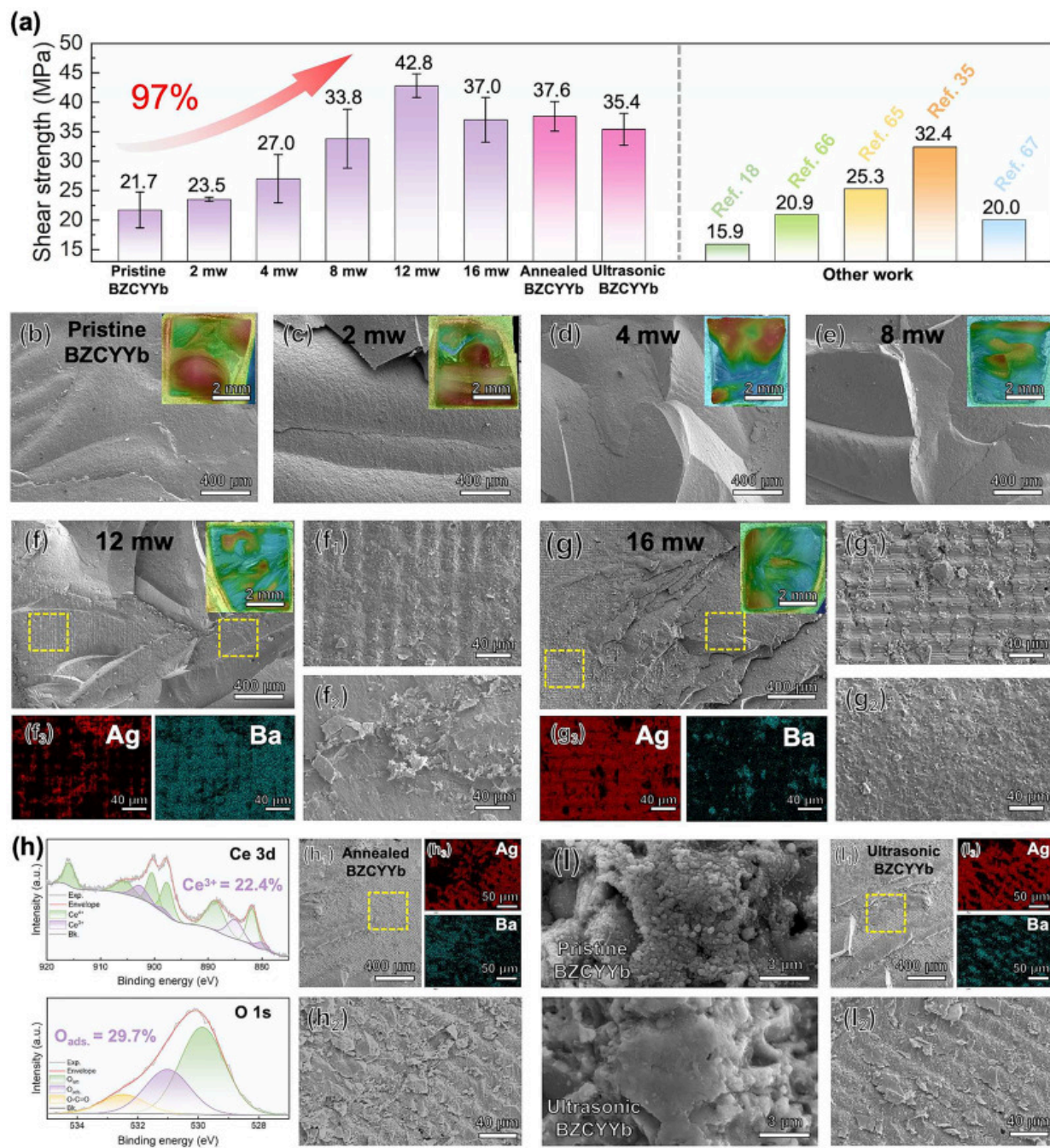
[Download: Download high-res image \(2MB\)](#)

[Download: Download full-size image](#)

Fig. 7. Microscopic morphology of the joints after exposure to 50% air-50% H₂O atmosphere at 600°C for different times. (a-c) Microscopic morphology of the joints after 100–300h aging; (a₁-c₁) Magnified view of Ag/stainless steel interface; (a₂-c₂) Magnified view of Ag/BZCYYb interface.

3.5. Mechanical properties and FEA calculations of BZCYYb/Crofer 22 APU joints

This work specifically addresses the mechanical joining and gas-tight sealing between the BZCYYb electrolyte and stainless-steel support frame, rather than the electrochemical reactions within the fuel cell itself. Although femtosecond laser surface grooving reduces the area-specific resistance (Fig. S6), the resistance at this joint region remains far higher than that of the cell electrolyte (typically $<1\Omega\cdot\text{cm}^2$). Consequently, almost no current conduction occurs across this joint during operation, and thus the joining process does not affect the electrochemical performance of the fuel cell. Instead, the mechanical performance of the joint is the decisive criterion for evaluating the effectiveness of this joining approach. [Fig. 8a](#) illustrates the influence of femtosecond laser grooving on the shear strength of BZCYYb/steel joints, in comparison with other joining techniques. The untreated joint exhibits a shear strength of 21.7MPa, which increases with rising laser power, reaching a peak of 42.8MPa at 12mW, exhibiting a 97% enhancement. However, the strength slightly decreases at 16mW due to partial disruption of the groove structure caused by excessive laser energy. The peak shear strength achieved in this study (42.8MPa) significantly exceeds that all previously reported BZCYYb/steel joining methods to our knowledge [[18,35,65](#)], [[66](#)], [[67](#)]].



Download: [Download high-res image \(3MB\)](#)

[Download: Download full-size image](#)

Fig. 8. Shear strength and fracture morphology of joints obtained with various laser processing powers (Crofer 22 APU side). (a) Shear strength of joints; fracture morphology of joints (b) without processing; (c) 2 mw; (d) 4 mw; (e) 8 mw; (f) 12 mw; (g) 16 mw; (h) annealed BZCYYb and (i) Ultrasonic BZCYYb.

Based on the earlier discussion that femtosecond laser modification introduces oxygen vacancies and nanoscale surface protrusions, which are expected to enhance interfacial joining, we further investigate their roles through selective removal experiments. For BZCYYb pretreated with 12mW femtosecond laser, a post-treatment at 1100°C for 2h removes surface oxygen vacancies, which has widely used for oxygen vacancies partially removal in oxide ceramics [[68], [69], [70]]. After annealing, the shear strength decreases to 37.6MPa. In addition, ultrasonic cleaning for 2h, previously reported to partially remove laser-induced nanoparticles [28], partially eliminates laser-induced surface nanoparticles, reducing the shear strength to 35.4MPa. It should be noted that it is very difficult to completely remove oxygen vacancies or nanoparticles solely by annealing or ultrasonic treatment without damaging the ceramic substrate, these results should be regarded as direct evidence that each factor affects shear strength, while the resulting shear strength reflects a partial influence of these factors.

To evaluate the influence of femtosecond laser processing on the fracture behavior of the BZCYYb/Ag interface, Fig. 8b-8g depict the fracture morphologies at various laser powers. In the untreated BZCYYb sample (Fig. 8b), cracks predominantly propagate within the ceramic matrix, indicating that residual stress remains concentrated within the ceramic. As laser power increases (Fig. 8c-8e), the crack path gradually shifts from the ceramic matrix to the interface. 3D morphological analysis confirms this transition, as the red regions (indicating internal ceramic fractures) diminish while the green and blue regions (representing interfacial fractures) expand, suggesting a growing dominance of interfacial failure. Previous studies have demonstrated that femtosecond laser-induced microstructures optimize residual stress distribution at the interface, effectively reducing

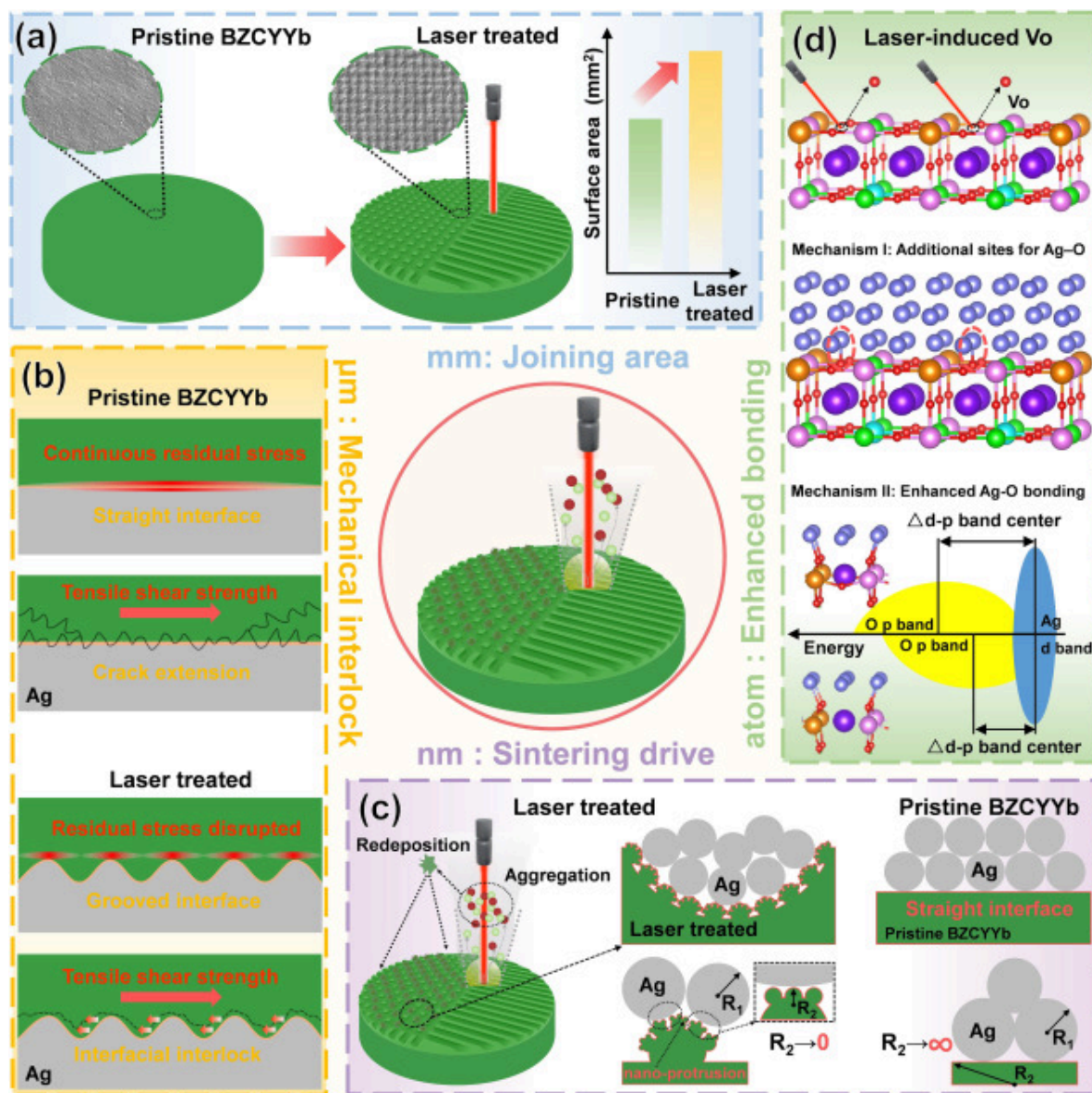
stress concentration [[23], [24], [25]]. At 12mW (Fig. 8f) and 16mW (Fig. 8g), fracture occurs almost entirely along the Ag/BZCYYb interface. Elemental mapping (Fig. 8f₃-8g₃) shows Ag and Ba signals on the fracture surfaces, confirming interfacial separation. Fig. 8f₁ reveals that Ag infiltrates ceramic grooves and leaves characteristic imprints after fracture. Fig. 8f₂ further shows ceramic fragments adhering to the Ag interlayer, indicating mechanical interlocking at the interface. This interfacial tearing requires additional energy, thereby enhancing shear resistance. Comparing Fig. 8f₂ and Fig. 8g₂, more ceramic residue is observed on the 12mW-treated joint, suggesting stronger interfacial engagement than at 16mW. Excessive laser energy at 16mW degrades groove morphology, weakening mechanical interlocking, consistent with the interface microstructure in Fig. 6f₃. Well-defined grooves deflect crack propagation and extend the crack path, improving overall joint toughness and shear performance.

As shown in Fig. 8h, after post-annealing treatment, the O_{ads} content decreases to 29.7%, accompanied by a reduction in the Ce³⁺ fraction to 22.4%, indicating a significant decrease in surface VOs. Despite this, fracture still occurs at the BZCYYb/Ag interface, as shown in Fig. 8h₁-8h₃. Fig. 8i reveals that ultrasonic cleaning effectively removes a substantial portion of the laser-induced nanoparticles compared to the untreated 12mW surface. However, the fracture likewise remains at the BZCYYb/Ag interface (Fig. 8i₁-i₃). These results suggest that although the grooved morphology formed by laser texturing may alleviate interfacial residual stress, the absence of oxygen vacancies and surface nanostructures diminishes interfacial joining, ultimately leading to reduced mechanical strength (Fig. 8a). In summary, at the microscale, the laser-induced grooves form mechanical interlocking at the interface that increases shear strength, as confirmed by the fracture morphology, while at the nanoscale and atomic scale, the nanoparticles and oxygen vacancies enhance interfacial sintering and bonding to increase shear strength, as verified by simulations presented later. The synergistic multiscale surface modification mechanism for strengthening shear performance is discussed in detail in the following section.

Due to the significant mismatch in thermal expansion coefficients between BZCYYb and Ag, considerable interfacial residual stresses develop during cooling. To investigate the regulatory effect of femtosecond laser-induced grooving on stress distribution at the BZCYYb/Ag interface, finite element simulations are conducted for both pristine and laser-treated joints during cooling from 930°C to room temperature (see Fig. S7 in the Supplementary information for details). As shown in Fig. S8a-S8b, the Mises and X-directional stress distributions reveal notable differences. In pristine joints, a continuous high-stress band forms along the interface and extends into the ceramic bulk, promoting interfacial crack initiation and propagation. In contrast, laser-grooved joints show localized and discontinuous stress concentrations near the grooves, along with a clear reduction in interfacial stress. This suggests that the grooves modify the stress transmission path and interrupt linear crack propagation. Fig. S8b shows that the pristine joint exhibits a sharp stress gradient and pronounced peak along the interface. After laser grooving, this concentration is significantly alleviated, resulting in a smoother stress gradient (Fig. S8b₁).

3.6. Formation mechanism of joints via multiscale interface regulation by femtosecond laser surface grooving

To enhance the strength of sintered joints between BZCYYb and Crofer 22 APU stainless steel, a multiscale interfacial strengthening strategy based on femtosecond laser surface structure modulation is proposed. As illustrated in Fig. 9, this strategy achieves synergistic strengthening from macro to atomic scales, effectively improving the mechanical properties of the joint. At the macroscopic scale (Fig. 9a), the periodic groove structure created by the femtosecond laser transforms the original flat interface into a three-dimensional morphology, resulting in a 44% increase in surface area. This significant increase in the interfacial joining area enhances the mechanical properties of the joint.



Download: [Download high-res image \(2MB\)](#)

Download: [Download full-size image](#)

Fig. 9. Multiscale interface regulation mechanisms enabled by femtosecond laser surface grooving for sintered joining of BZCYYb and Crofer 22 APU. (a) Surface topography

evolution induced by laser grooving. (b) Redistribution of residual stress and interfacial mechanical interlocking. (c) Nanostructure-induced sintering enhancement via curvature regulation. (d) Atomic-scale bonding mechanisms facilitated by oxygen vacancies.

At the microscale (Fig. 9b), periodic microgrooves interrupt the continuity of residual stress fields and promote mechanical interlocking through meshing effects. During fracture, this meshing action induces crack deflection and path elongation, thereby enhancing mechanical properties of the joint. Notably, this mechanism aligns with the findings of Zhang et al. [23]. This mechanism will be elaborated in detail in the subsequent section on fracture analysis. Bi et al. [25] also highlighted that femtosecond laser grooves can improve joining stability by redistributing interfacial stresses.

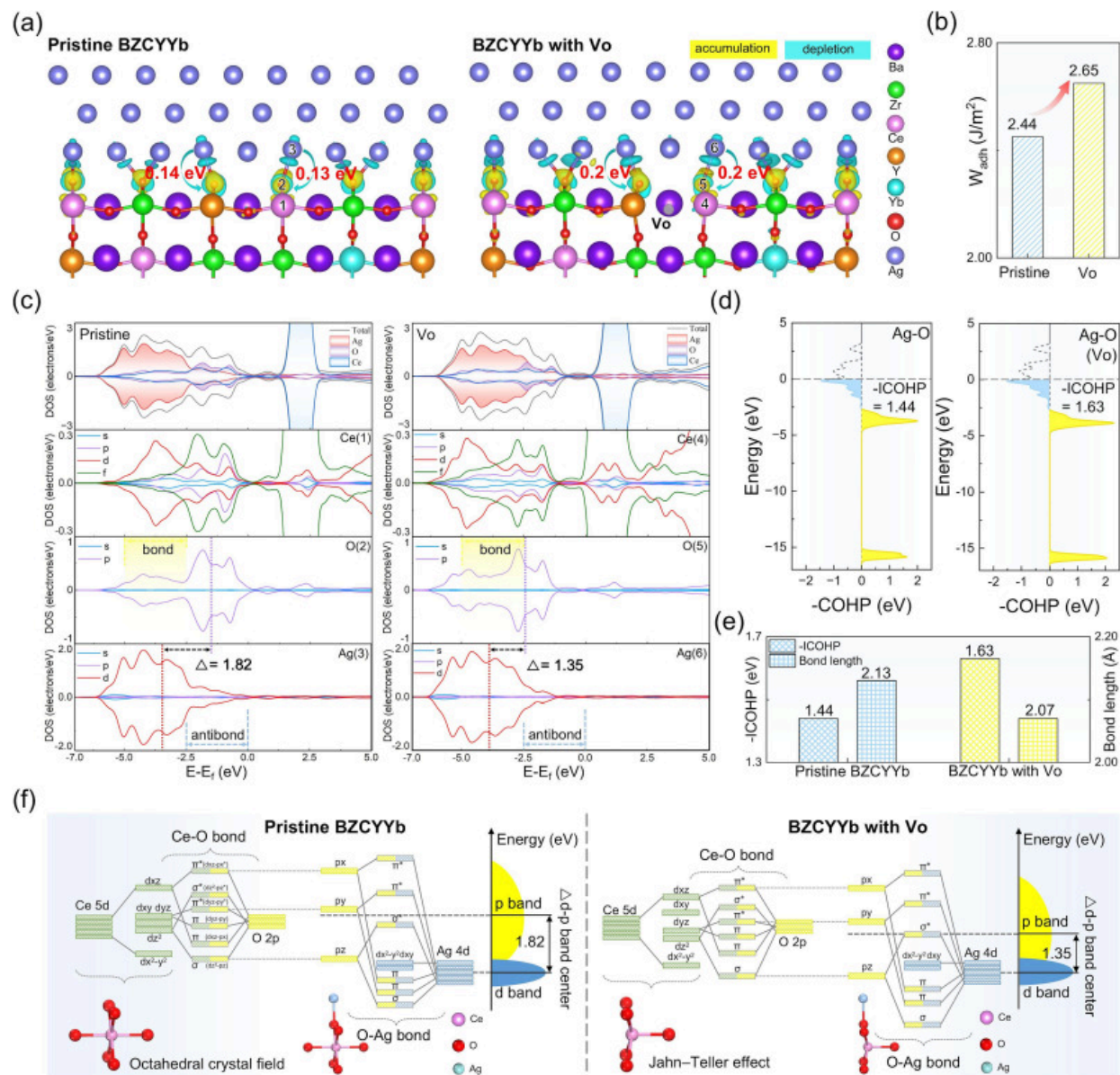
At the nanoscale (Fig. 9c), the high-temperature plasma plume generated by laser ablation produces numerous nanoparticles with elevated surface energy during the processes of diffusion and redeposition. The relationship between laser-redeposited particles and the regulation of micro and nanostructures has been systematically elucidated by Long et al. [27,28]. In the context of interfacial sintering, the Laplace equation is commonly used to describe the sintering driving force, as illustrated in the following Eq. [71]:

$$P = \gamma \left(\frac{1}{R_1} + \frac{1}{R_2} \right) + P_0 \quad (2)$$

Where P represents the driving force at the interface, P_0 is the external pressure on the surface, and γ denotes the surface energy. This surface energy depends on both the surface stress derivation and the surface chemical contributions of the particles, which can be calculated using the equation $\gamma = (\partial G / \partial A)_{T,P,n}$ [29]. The surface energy γ is characterized as an excess free energy, and the reduction in surface area facilitates the joining process [72]. R_1 and R_2 denote the radii of curvature of the particles at the contact points of the surface. Given that the surface of pristine BZCYYb approaches flatness ($R_2 \rightarrow \infty$), the sintering driving force is primarily influenced by the curvature of the Ag

particles (R_1). The laser-induced nanostructures significantly reduce R_2 ($R_2 \rightarrow 0$), thereby enhancing the sintering driving force.

At the atomic scale (Fig. 9d), femtosecond laser irradiation induces oxygen vacancies within the BZCYYb lattice, as validated in this study. Oxygen vacancies improve the Ag/BZCYYb interfacial joining strength via two mechanisms. Existing studies have shown that O vacancies at the interface provide additional sites for Ag—O bonding, thereby enhancing the interfacial adhesion work (W_{adh}) [33,34]. However, this mechanism is contingent upon the oxidation of Ag at the interface and has limitations in explaining profound changes in electronic structure. In this study, we propose that oxygen vacancies induce crystal field reconstruction, thereby enhancing d-p orbital coupling. The formation of oxygen vacancies disrupts the octahedral coordination symmetry of B-site cations (e.g. Ce), leading to crystal field splitting and orbital energy rearrangement. This shift lowers the energy of the coordinated O p orbitals, thereby reducing the energy difference (ΔE_{d-p}) between the O 2p band center and the Ag 4d band center. The reduced ΔE_{d-p} enhances d-p orbital hybridization between the Ag 4d and O 2p orbitals, thereby strengthening interfacial covalent bonding. A more detailed discussion of this mechanism is presented in the section on first-principles calculations (Fig. 10).



[Download: Download high-res image \(2MB\)](#)

[Download: Download full-size image](#)

Fig. 10. Atomic-scale mechanisms of oxygen vacancy-enhanced interfacial bonding at the Ag/BZCYYb interface. (a) Charge redistribution and Bader charge analysis near oxygen

vacancies. (b) Quantification of interfacial adhesion energy. (c) PDOS comparison for Ce-O-Ag triplets with and without oxygen vacancies. (d) COHP analysis of Ag—O bonding strength. (e) Variation of -ICOHP values and Ag—O bond length. (f) Orbital-level interpretation based on crystal field theory and Jahn-Teller effects.

3.7. First-principles calculations on oxygen vacancy-enhanced interfacial bonding

Fig. 10a illustrates the distribution of the electron differential density and Bader charge before and after oxygen vacancy formation at the Ag/BZCYYb interface. Enhanced electron accumulation (blue region) above Ag atoms near oxygen vacancy sites indicates increased charge transfer and stronger Ag-ceramic electronic coupling. Quantitatively, Fig. 10b shows that the interfacial adhesion work (W_{adh}) increases from 2.44 to 2.65 J/m² with oxygen vacancy, confirming improved interfacial bonding.

To investigate the mechanism behind this enhancement, the Ce-O-Ag triatomic configuration is selected as a representative interfacial unit. Partial density of states (PDOS) analyses is conducted for two triplets: Ce(1)/O(2)/Ag(3) without oxygen vacancy and Ce(4)/O(5)/Ag(6) with oxygen vacancy (Fig. 10c). In both cases, Ce 5d orbitals hybridize with O 2p orbitals between -5 and 0 eV. In the presence of oxygen vacancy, crystal field distortion induces orbital splitting of the Ce 5d and 4f orbitals, shifting them to lower energies. The emergence of additional peaks reflects a breakdown of orbital degeneracy due to the Jahn-Teller distortion, consistent with the transition from a symmetric to an asymmetric local crystal field [73,74]. For Ag—O interactions, the crystal orbital Hamilton population (COHP) analysis (Fig. 9d) shows a clear division of the state density into bonding (-5 to -2.5 eV) and antibonding (-2.5 to 0 eV) regions. In the bonding region, the overlap between Ag 4d and O 2p orbitals increases significantly in the presence of oxygen vacancy. This enhancement is evidenced by an increase in the integrated COHP (-ICOHP) value from 1.44 eV (without oxygen vacancy) to 1.63 eV (with oxygen vacancy), as visually indicated by the expansion of the yellow bonding region in the -5 to -2.5 eV range.

This enhancement indicates enhanced covalent interactions between Ag and O. The shortening of the Ag—O bond length from 2.13 Å to 2.07 Å (Fig. 10e) further confirms this trend. Additionally, the energy level difference (ΔE_{d-p}) between the O 2p band center and the Ag 4d band center is calculated. Upon oxygen vacancy formation, the O 2p band center shifts from -1.65 eV to -2.49 eV, while the Ag 4d band center shifts from -3.48 eV to -3.85 eV. As a result, this reduced ΔE enhances d-p orbital hybridization, strengthening chemical bonding at the interface [75,76].

A deeper interpretation of orbital behavior is provided in Fig. 10f based on crystal field theory. In the absence of oxygen vacancy, the Ce atom is octahedrally coordinated by oxygen. The Ce 5d orbital splitting pattern can be derived through the evolution of the classical t_{2g} -eg pattern, as further elaborated in the supplementary material (Fig. S9) [77]. According to symmetry selection rules, the Ce d_{z^2} orbital engages in σ -type bonding and antibonding interactions with the O p_z orbital, while the dxz and dyz orbitals form π -type bonds with the O p_x and p_y orbitals, respectively. The O 2p orbitals also participate in similar σ - π bonding interactions with the Ag 4d orbitals [78]. When oxygen vacancy formation, the absence of a ligand weakens electrostatic repulsion, resulting in a downward shift of the Ce dx^2-y^2 , dxz , and dz^2 orbitals and undergo desymmetrization [79]. This corresponds to the appearance of new low-energy peaks in the PDOS and detailed further in the supplementary material (Fig. S10). These orbitals subsequently hybridize with the O 2p orbitals to form lower-energy bonding states, leading to a downward shift in the O 2p band center and improved energy alignment with the Ag 4d band center. Such alignment promotes enhanced d-p orbital coupling and reinforces the covalent nature of the Ag—O bond. This enhancement is evidenced by the increased overlap between Ag 4d and O 2p states in the -5 to -2.5 eV range in Fig. 9c and the expanded yellow bonding region in Fig. 9d, contributing to increased interfacial W_{adh} .

4. Conclusions

This study proposes an efficient ceramic-metal joining strategy that enables dense and stable sintered bonding between BZCYYb ceramics and Crofer 22 APU metal through

multiscale interface regulation induced by femtosecond laser structuring, synergized with a micro-Ag particle interlayer. The main conclusions are as follows:

- (1) Femtosecond laser processing induces a hierarchical evolution of the ceramic surface, introducing micron-scale grooves ($15\mu\text{m}$ wide and $2.5\mu\text{m}$ deep), nanoscale clusters (ranging from several tens to hundreds of nanometers), and atomic-scale oxygen vacancies, with vacancy concentration increased by 100% at a laser power of 12mW. This well-defined multiscale architecture significantly enhances sintering kinetics and interfacial bonding capability.
- (2) Under optimized processing parameters (930°C , 1MPa, 30min, and 12mW laser power), Ag fully fills the laser-induced grooves on the ceramic surface, forming a tightly joined interface dominated by diffusion without any brittle reaction layers. On the metal side, a continuous and dense oxide layer ($\sim 2\mu\text{m}$) forms. The resulting joints exhibit excellent mechanical and chemical stability after aging for up to 300h in a humid oxidative atmosphere.
- (3) The joints exhibit outstanding mechanical performance, with shear strength reaching 42.8MPa, representing a 97% improvement over untreated samples. Fracture morphology and FEA reveal that the groove structure effectively redistributes residual at the interface and suppresses crack propagation.
- (4) The joining behavior exhibits multiscale synergy, spanning an enlarged macroscopic joining area, microscale mechanical interlocking, nanoscale sintering driving force, and electronic structure modulation. First-principles calculations reveal that laser-induced oxygen vacancies trigger Jahn-Teller distortion in the crystal field, enhancing d-p orbital hybridization between coordinating O and Ag. This interaction strengthens interfacial covalent

bonding (-ICOHP increases from 1.44eV to 1.63eV) and improves adhesion (W_{adh} increases from 2.44J/m² to 2.65J/m²).

This work offers a scalable and practical pathway for fabricating robust ceramic-metal interfaces, with promising implications for advanced energy systems and high-performance electrochemical devices.

CRediT authorship contribution statement

Naiqiang Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Xiaoqing Si:** Writing – review & editing, Validation, Supervision, Methodology, Data curation, Conceptualization. **Hanyue Ding:** Writing – review & editing, Methodology, Data curation. **Zhihu Zhang:** Writing – review & editing, Methodology, Data curation. **Jianwei Gao:** Writing – review & editing, Methodology, Data curation. **Shaletanati Ake:** Writing – review & editing, Methodology, Data curation. **Xuexue Li:** Writing – review & editing, Methodology, Data curation. **Chun Li:** Writing – review & editing. **Junlei Qi:** Writing – review & editing. **Jian Cao:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors gratefully acknowledge the financial support from the National Natural Science Foundation of China [grant numbers [52375323](#) and [52125502](#)], and the China Postdoctoral Science Foundation [2019TQ0075](#). Central Universities [grant numbers [FRFCU5710051121](#) and [FRFCU5720100421](#)].

Appendix A. Supplementary data

 [Download: Download Word document \(4MB\)](#)

Supplementary material

Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study.

References

- [1] F. Liu, H. Deng, D. Diercks, P. Kumar, M.H.A. Jabbar, C. Gumezi
Duan C, lowering the operating temperature of protonic ceramic
electrochemical cells to < 450 °C
Nat. Energy, 8 (10) (2023), pp. 1145-1157
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [2] Y. Hu, D. Li, H. Guo
Recent progress of high-performance interconnectors for SOFC: from
materials, protective coatings, optimizing strategies, towards the real
stack applications
Chem. Eng. J. (2025), p. 159321
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [3] C. Duan, J. Tong, M. Shang, S. Nikodemski, M. Sanders, S. Ricote, R. Hayre
Readily processed protonic ceramic fuel cells with high performance at
low temperatures
Science, 349 (6254) (2015), pp. 1321-1326
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [4] L. Yang, S. Wang, K. Blinn, M. Liu, Z. Liu, Z. Cheng, M.L. Liu
Enhanced sulfur and coking tolerance of a mixed ion conductor for SOFCs:
 $\text{BaZr}_{0.1}\text{Ce}_{0.7}\text{Y}_{0.2-x}\text{Yb}_x\text{O}_{3-\delta}$
Science, 326 (5949) (2009), pp. 126-129
[Crossref ↗](#) [Google Scholar ↗](#)
- [5] S. Choi, C.J. Kucharczyk, Y. Liang, X. Zhang, I. Takeuchi, H.I. Ji, S.M. Haile
Exceptional power density and stability at intermediate temperatures in
protonic ceramic fuel cells
Nat. Energy, 3 (3) (2018), pp. 202-210
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [6] B. Wu, X. Yu, Z. Zhao
Improving the catalytic activity and sintering resistance of Ni-
 $\text{BaCe}_{0.7}\text{Zr}_{0.1}\text{Y}_{0.1}\text{Yb}_{0.1}\text{O}_{3-\delta}$ cermet anode for ammonia-fueled protonic
ceramic fuel cells via cobalt addition
Chem. Eng. J., 507 (2025), Article 160757
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [7] C. Duan, R.J. Kee, H. Zhu
Highly durable, coking and sulfur tolerant, fuel-flexible protonic ceramic
fuel cells
Nature, 557 (7704) (2018), pp. 217-222
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [8] F. Ramadhani, M.A. Hussain, H. Mokhlis
Optimization strategies for solid oxide fuel cell (SOFC) application: a
literature survey
Renew. Sust. Energ. Rev., 76 (2017), pp. 460-484
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [9] Y. Zhao, S. Zhang, M. Su

In-situ formed CuFe_2O_4 spinel coating by electroplating method for solid oxide fuel cell interconnect

Chem. Eng. J., 470 (2023), Article 144397



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[10]

A.G. Sabato, E. Zanchi, S. Molin

Mn-co spinel coatings on Crofer 22 APU by electrophoretic deposition: up scaling, performance in SOFC stack at 850 °C and compositional modifications

J. Eur. Ceram. Soc., 41 (8) (2021), pp. 4496-4504



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[11]

R. Wang, Z. Sun, J.P. Choi

Ferritic stainless steel interconnects for protonic ceramic electrochemical cell stacks: oxidation behavior and protective coatings

Int. J. Hydrog. Energy, 44 (47) (2019), pp. 25297-25309



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[12]

W.Z. Zhu, S.C. Deevi

Development of interconnect materials for solid oxide fuel cells

Mater. Sci. Eng. A, 348 (1–2) (2003), pp. 227-243



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[13]

R. Kiebach, K. Engelbrecht, L. Grahl-Madsen, *et al.*

An ag based brazing system with a tunable thermal expansion for the use as sealant for solid oxide cells

J. Power Sources, 315 (2016), pp. 339-350



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[14]

B. Kuhn, F.J. Wetzel, J. Malzbender, *et al.*

Mechanical performance of reactive-air-brazed (RAB) ceramic/metal joints for solid oxide fuel cells at ambient temperature

J. Power Sources, 193 (1) (2009), pp. 199-202



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[15]

K.S. Weil, J.Y. Kim, J.S. Hardy

Reactive air brazing: a novel method of sealing SOFCs and other solid-state electrochemical devices

Electrochem. Solid-State Lett., 8 (2) (2005), p. A133

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[16]

R. Chatzimichail, G. Triantafyllou, F. Tietz, *et al.*

Interfacial properties of (ag+ CuO) brazes used as sealing materials in SOFC stacks

J. Mater. Sci., 49 (1) (2014), pp. 300-313

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[17]

Z. Huang, Y. Yang, H. Lv

Large-area anode-supported protonic ceramic fuel cells combining with multilayer-tape casting and hot-pressing lamination technology

J. Eur. Ceram. Soc., 43 (2) (2023), pp. 428-437



[View PDF](#) [View article](#) [Google Scholar ↗](#)

[18]

X. Wang, C. Li, Q. Zhou

Silver particle interlayer with high dislocation density for improving the joining of BaZr_{0.1}Ce_{0.7}Y_{0.1}Yb_{0.1}O_{3-δ} electrolyte and AISI 441 interconnect

J. Mater., 8 (5) (2022), pp. 1001-1008



[View PDF](#) [View article](#) [Crossref ↗](#) [Google Scholar ↗](#)

[19]

Z. Zhao, S. Janasekaran, G.T. Fong, *et al.*

Laser applications in ceramic and metal joining: a review

Met. Mater. Int., 30 (7) (2024), pp. 1743-1782

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [20] K. Sugioka, Y. Cheng
Ultrafast lasers—reliable tools for advanced materials processing
Light Sci Appl, 3 (4) (2014), Article e149-e149
[Google Scholar ↗](#)
- [21] Y. Song, H. Zhu, D. Liu
Synergistically enhanced Si₃N₄/Cu heterostructure bonding by laser surface modification
J. Mater. Sci. Technol., 182 (2024), pp. 187-197
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [22] J. Cao, Y. Song, H. Zhu
Nanosecond laser-induced surface modification to strengthen Cu/Si₃N₄ AMB joints
Mater. Sci. Eng. A, 850 (2022), Article 143573
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [23] Y. Zhang, G. Zou, L. Liu
Vacuum brazing of alumina to stainless steel using femtosecond laser patterned periodic surface structure
Mater. Sci. Eng. A, 662 (2016), pp. 178-184
 [View PDF](#) [View article](#) [Crossref ↗](#) [Google Scholar ↗](#)
- [24] W. Guo, T. Gao, X. Cui
Interfacial reactions and zigzag groove strengthening of C/C composite and Rene N5 single crystal brazed joint
Ceram. Int., 41 (9) (2015), pp. 11605-11610
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [25] X. Bi, Z. Wang, M. Xu

Femtosecond laser fabricated micro/nano interfacial structures to strengthen CFRPEEK/A6061-T6 FLJ hybrid joints

Compos. Part B Eng., 231 (2022), Article 109540



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[26] M.F. Ashby, H. Shercliff, D. Cebon

Chapter 13-diffusion and creep: materials at high temperatures

Materials: engineering, Science, Processing and Design, Butterworth-Heinemann (2018),

pp. 335-377

[Google Scholar ↗](#)

[27] J. Long, Z. Cao, C. Lin

Formation mechanism of hierarchical Micro-and nanostructures on copper induced by low-cost nanosecond lasers

Appl. Surf. Sci., 464 (2019), pp. 412-421



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[28] J. Long, Q. Peng, G. Chen

Centimeter-scale low-damage micromachining on single-crystal 4H-SiC substrates using a femtosecond laser with square-shaped flat-top focus spots

Ceram. Int., 47 (16) (2021), pp. 23134-23143



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[29] L. Vitos, A.V. Ruban, H.L. Skriver

The surface energy of metals

Surf. Sci., 411 (1–2) (1998), pp. 186-202



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[30] Liang Misheng, Xin Li, Lan Jiang, Peng Ran, Hongzhi Wang

Femtosecond laser mediated fabrication of micro/nanostructured TiO_{2-x} photoelectrodes: hierarchical nanotubes array with oxygen vacancies and

their photocatalysis properties

Appl. Catal. B Environ., 277 (2020), Article 119231

[Google Scholar ↗](#)

- [31] Y. Lu, C. Li, Y. Zhang
Engineering of cation and anion vacancies in Co_3O_4 thin nanosheets by laser irradiation for more advancement of oxygen evolution reaction
Nano Energy, 83 (2021), Article 105800
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [32] Z. Ning, Y. Lian, L. Jiang
Femtosecond laser-induced anisotropic structure and nonlinear optical response of yttria-stabilized zirconia single crystals with different planes
ACS Appl. Mater. Interfaces, 14 (34) (2022), pp. 39591-39600
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [33] T. Phongpreecha, J.D. Nicholas, T.R. Bieler
Computational design of metal oxides to enhance the wetting and adhesion of silver-based brazes on yttria-stabilized-zirconia
Acta Mater., 152 (2018), pp. 229-238
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [34] N. Wang, X. Si, B. Zhang
Micro-ag particle interlayer with high dislocation density for sintered-joining of Al_2O_3 and AlN in air
Ceram. Int., 50 (17) (2024), pp. 29153-29168
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [35] X. Wang, X. Si, M. Li
 $\text{Y}_2\text{W}_3\text{O}_{12}@\text{SiO}_2$ composite particles for regulating thermal expansion and interfacial reactions in $\text{BaZr}_{0.1}\text{Ce}_{0.7}\text{Y}_{0.1}\text{Yb}_{0.1}\text{O}_{3-\delta}$ /AISI 441 joints
Compos. Part B Eng., 242 (2022), Article 110108

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [36] G. Kresse, J. Furthmüller
Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set

Phys. Rev. B, 54 (16) (1996), p. 11169

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [37] J.P. Perdew, K. Burke, M. Ernzerhof
Generalized gradient approximation made simple

Phys. Rev. Lett., 77 (18) (1996), p. 3865

[Google Scholar ↗](#)

- [38] S. Grimme, J. Antony, S. Ehrlich
A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H-Pu

J. Chem. Phys., 132 (15) (2010)

[Google Scholar ↗](#)

- [39] V.I. Anisimov, J. Zaanen, O.K. Andersen
Band theory and Mott insulators: Hubbard U instead of stoner I

Phys. Rev. B, 44 (3) (1991), p. 943

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [40] Q. Zhang, X. Zheng, J. Jiang
Structural stability of $\text{La}_2\text{Ce}_2\text{O}_7$ as a proton conductor: a first-principles study

J. Phys. Chem. C, 117 (40) (2013), pp. 20379-20386

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [41] G. Zhang, G. Chen, C. Panwisawas

Uncovering the fracture mechanism of laves (111)/Ni₆Nb₇ (0001) interfaces by first-principles calculations

Acta Mater., 281 (2024), Article 120426



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [42] S.K. Sundaram, E. Mazur
Inducing and probing non-thermal transitions in semiconductors using femtosecond laser pulses

Nat. Mater., 1 (4) (2002), pp. 217-224

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [43] H. Palneedi, J.H. Park, D. Maurya
Laser irradiation of metal oxide films and nanostructures: applications and advances

Adv. Mater., 30 (14) (2018), p. 1705148

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [44] X. Yang, L. Jia, B. Pan
Mechanism of proton conduction in doped barium cerates: a first-principles study

J. Phys. Chem. C, 124 (14) (2020), pp. 8024-8033

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [45] R. Le Harzic, N. Huot, E. Audouard
Comparison of heat-affected zones due to nanosecond and femtosecond laser pulses using transmission electronic microscopy

Appl. Phys. Lett., 80 (21) (2002), pp. 3886-3888

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [46] C.Y. Shih, M.V. Shugayev, C. Wu

Generation of subsurface voids, incubation effect, and formation of nanoparticles in short pulse laser interactions with bulk metal targets in liquid: molecular dynamics study

J. Phys. Chem. C, 121 (30) (2017), pp. 16549-16567

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [47] S. Imashuku, T. Uda, Y. Awakura
Sintering properties of trivalent cation-doped barium zirconate at 1600° C

Electrochem. Solid-State Lett., 10 (10) (2007), p. B175

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [48] Y. Yamazaki, R. Hernandez-Sanchez, S.M. Haile
Cation non-stoichiometry in yttrium-doped barium zirconate: phase behavior, microstructure, and proton conductivity

J. Mater. Chem., 20 (37) (2010), pp. 8158-8166

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [49] Y. Song, H. Zhu, D. Liu
Direct bonding of silicon nitride to copper via laser surface modification

Appl. Surf. Sci., 602 (2022), Article 154354



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [50] Y. Zhao, Y. Zhu, J. Zhu
Atomic-resolution investigation of structural transformation caused by oxygen vacancy in $\text{La}_{0.9}\text{Sr}_{0.1}\text{TiO}_{3+\delta}$ titanate layer perovskite ceramics

J. Mater. Sci. Technol., 104 (2022), pp. 172-182



[View PDF](#) [View article](#) [Google Scholar ↗](#)

- [51] K. Zhu, F. Shi, X. Zhu
The roles of oxygen vacancies in electrocatalytic oxygen evolution reaction

Nano Energy, 73 (2020), Article 104761

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [52] M. Zhu, K. Liu, Y. Meng
Enhanced temperature stability of colossal-permittivity Ce-doped SrTiO₃ ceramics designed by defect engineering

J. Am. Ceram. Soc., 108 (2025), Article e20365

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [53] V.V. Atuchin, T.A. Gavrilova, J.C. Grivel
Electronic structure of layered ferroelectric high-k titanate Pr₂Ti₂O₇

J. Solid State Chem., 195 (2012), pp. 125-131

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [54] Y.T. Tsai, W.J. Chang, S.W. Huang
Temperature-dependent X-ray absorption near edge spectroscopy of doped LaMnO₃: ion-size and the Jahn-teller distortion effects

Phys. B Condens. Matter, 404 (8–11) (2009), pp. 1404-1408

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [55] Q. Ren, Y. He, H. Wang
Photo-switchable oxygen vacancy as the dynamic active site in the photocatalytic NO oxidation reaction

ACS Catal., 12 (22) (2022), pp. 14015-14025

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [56] X. Hao, A. Yoko, K. Inoue
Atomistic origin of high-concentration Ce³⁺ in {100}-faceted Cr-substituted CeO₂ nanocrystals

Acta Mater., 203 (2021), Article 116473

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [57] G. Chen, Y. Luo, W. Sun

Electrochemical performance of a new structured low temperature SOFC with BZY electrolyte

Int. J. Hydrog. Energy, 43 (28) (2018), pp. 12765-12772



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [58] S. Wang, J. Zhu, T. Li
Oxygen vacancy-mediated CuCoFe/tartrate-LDH catalyst directly activates oxygen to produce superoxide radicals: transformation of active species and implication for nitrobenzene degradation

Environ. Sci. Technol., 56 (12) (2022), pp. 7924-7934

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [59] G. Hu, T.P. Hogan, J.D. Nicholas
Porous interlayers that getter surface-segregating species for improved silver wetting, adhesion, and electrical contact on stainless steel SOFC components

J. Electrochem. Soc., 172 (2025), Article 014513

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [60] H. Chen, L. Li, R. Kemps
Reactive air brazing for sealing mixed ionic electronic conducting hollow fibre membranes

Acta Mater., 88 (2015), pp. 74-82



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [61] J. Cao, Z. Wang, C. Li
Microstructure evolution and mechanical properties of co coated AISI 441 ferritic stainless steel/YSZ reactive air brazed joint

Int. J. Hydrog. Energy, 46 (12) (2021), pp. 8758-8766



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [62] K. Lee, B. Yoon, J. Kang
Evaluation of ag-doped $(\text{MnCo})_3\text{O}_4$ spinel as a solid oxide fuel cell metallic interconnect coating material
Int. J. Hydrog. Energy, 42 (49) (2017), pp. 29511-29517
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [63] Z. Zhang, C. Chen, A. Suetake
Pressureless and low-temperature sinter-joining on bare Si, SiC and GaN by a Ag flake paste
Scr. Mater., 198 (2021), Article 113833
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [64] M. Chen, K. Zhao, J. Zhao
Accelerated kinetics of hydrogen oxidation reaction on the Ni anode coupled with $\text{BaZr}_{0.9}\text{Y}_{0.1}\text{O}_{3-\delta}$ proton-conducting ceramic electrolyte via tuning the electrolyte surface chemistry
Appl. Surf. Sci., 605 (2022), Article 154800
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [65] X. Wang, X. Si, C. Li
Joining the $\text{BaZr}_{0.1}\text{Ce}_{0.7}\text{Y}_{0.1}\text{Yb}_{0.1}\text{O}_{3-\delta}$ electrolyte to AISI 441 interconnect for protonic ceramic fuel cell applications: interfacial microstructure and long-term stability
ACS Appl Energy Mater, 4 (7) (2021), pp. 7346-7354
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [66] X. Wang, X. Si, J. Gao
Effects of the Mn-co spinel coating on $\text{BaZr}_{0.1}\text{Ce}_{0.7}\text{Y}_{0.1}\text{Yb}_{0.1}\text{O}_{3-\delta}$ /AISI 441 reactive air brazed joint: microstructure, strength, and stability
Ceram. Int., 49 (3) (2023), pp. 5154-5160
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [67] J. Gao, X. Si, B. Yang
TiO₂ containing glass sealant for protonic ceramic fuel cell applications: microstructure, strength and stability studied in sealing of BaZr_{0.1}Ce_{0.7}Y_{0.1}Yb_{0.1}O_{3-δ} electrolyte to Crofer 22H interconnect
Ceram. Int., 50 (23) (2024), pp. 51207-51217
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [68] K. Kim, C. Lee, W.Y. Lee, *et al.*
Enhanced switching ratio of sol-gel-processed Y₂O₃ RRAM device by suppressing oxygen vacancy formation at high annealing temperatures
Semicond. Sci. Technol., 37 (1) (2021), p. 015007
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [69] V. Khandelwal, P. Siroha, S. Barik, *et al.*
Impact of oxygen vacancy reduction on the dielectric, energy storage, and electrocaloric properties of annealed BCZT ceramic
J. Mater. Chem. C, 13 (20) (2025), pp. 10178-10193
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [70] D. Petrova, B. Napoleonov, C.N.H. Minh, *et al.*
The effect of post deposition treatment on properties of ALD Al-doped ZnO films
Nanomaterials, 13 (5) (2023), Article 800
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [71] R.M. German
Thermodynamics of Sintering
Sintering of Advanced Materials, Elsevier (2010), pp. 3-32
 [View PDF](#) [View article](#) [Crossref ↗](#) [Google Scholar ↗](#)
- [72] G. Ouyang, X. Tan, G. Yang
Thermodynamic model of the surface energy of nanocrystals

Phys. Rev. B-Condensed Matter and Materials Physics, 74 (19) (2006), Article 195408

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [73] I. Živković, J.R. Soh, O. Malanyuk
Dynamic Jahn-Teller effect in the strong spin-orbit coupling regime

Nat. Commun., 15 (1) (2024), Article 8587

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [74] I. Mosquera-Lois, S.R. Kavanagh, a Walsh
Identifying the ground state structures of point defects in solids

NPJ Comput. Mater., 9 (1) (2023), Article 25

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [75] J. Zhou, X. Liu, L. Zhu
Deciphering the modulation essence of p bands in co-based compounds
on Li-S chemistry

Joule, 2 (12) (2018), pp. 2681-2693



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [76] Z. Li, X. Chen, G. Yao
Strengthening d-p orbital-hybridization via coordination number
regulation of manganese single-atom catalysts toward fast kinetic and
Long-life sodium-sulfur batteries

Adv. Funct. Mater., 34 (34) (2024), p. 2400859

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [77] B.G. Diamond, L.I. Payne, C.H. Hendon
Ligand field tuning of d-orbital energies in metal-organic framework
clusters

Communications Chemistry, 6 (1) (2023), p. 67

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [78] X. Liang, X. Wu, J. Hu
Large magnetic anisotropy in chemically engineered iridium dimer

Commun. Phys., 1 (1) (2018), p. 74



[View PDF](#)

[View article](#)

[Google Scholar ↗](#)

- [79] W.J. Kim, M.A. Smeaton, C. Jia
Geometric frustration of Jahn-teller order in the infinite-layer lattice

Nature, 615 (7951) (2023), pp. 237-243

[Crossref ↗](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[View Abstract](#)

Recommended articles

Based on reading popularity

Short communication Full text access

A novel brush surface structure of SiC_f/SiC composites designed for brazing improvement

Jin Ba, ..., Junlei Qi

Vacuum • Volume 195 • 2022 • Article 110700



[View PDF](#)

Research article Full text access

Enhanced bonding of Al₂O₃/Al₂O₃ joints brazed by Ni₅₀Ti₅₀ master alloy interlayer

Qi Zhang, ..., Hongwang Yang

Vacuum • Volume 185 • 2021 • Article 110000



[View PDF](#)

Research article Full text access

Forming and weld grain growth behavior of arc stud welding driven by rotating magnetic field

Deku Zhang, ..., Kehong Wang

Journal of Manufacturing Processes • Volume 82 • 2022 • Pages 138-151



[View PDF](#)

Research article Full text access

Brazing Al₂O₃ to 4J42 using Ag–Cu–Ti/Cu/BNi-2 composite fillers with different thicknesses of Cu interlayer

Qianying Zhu, ..., Yong Zhang

Ceramics International • Volume 49, Issue 6 • 2023 • Pages 9779-9788



[View PDF](#)

Research article Open access

The characteristics and dynamics of fused silica-aluminum alloy welding during mJ-level femtosecond laser

Lin Zhang, ..., Xiuquan Ma

Materials & Design • Volume 239 • 2024 • Article 112790



[View PDF](#)

Research article Full text access

SiC-7A52 brazed joint reinforced by femtosecond laser surface modification

Deku Zhang, ..., Xiaopeng Li

Journal of Manufacturing Processes • Volume 125 • 2024 • Pages 442-455



[View PDF](#)

Metrics

Captures

Mendeley Readers

6



[View details](#) ↗

All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.